HAWAI‘I WRRC
University of Hawai‘i at Mānoa

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REPORT TO THE HAWAI‘I STATE LEGISLATURE

Development of the Act 217 Sewer Expansion Overlay of the Hawai‘i Cesspool Prioritization Tool

Project Findings

Prepared in response to an August 10, 2026 request from the Speaker of the Hawai‘i House of Representatives and the Chair of the House Committee on Energy & Environmental Protection regarding findings from work conducted pursuant to Act 217, Session Laws of Hawai‘i 2024.

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Executive Summary

Act 217 directed the University of Hawai‘i Water Resources Research Center to identify where centralized sewer expansion may be most technically and economically feasible as a cesspool replacement strategy. This report presents the results of that analysis and describes the Act 217 Sewer Expansion Overlay Tool, a publicly accessible web application developed to communicate findings statewide. This new map and additional information are available on the Hawai‘i Cesspool Prioritization Tool site at www.hawaiicesspooltool.org. This study identifies 103 Potential Project Areas across five islands, covering approximately 39,300 cesspools, roughly half the statewide inventory. These areas represent the strongest candidate locations where cesspool density makes municipal-scale gravity sewer possible.

Key Takeaways

- **Feasibility and Implementation:** Centralized sewer expansion is substantially more expensive than on-site individual wastewater conversion in almost every situation this study examines. Realistic near-term candidates are limited to infill areas where costs are lowest and existing infrastructure is close. Beyond those, sewer expansion will require either exceptional environmental justification or levels of public investment that have not yet materialized. For the majority of the state’s cesspools, conventional sewer is unlikely to serve as the primary replacement pathway for the 2050 mandate.
- **Planning and Implementation Progress Gap:** This study identifies 103 Potential Project Areas encompassing 39,311 cesspools. Currently, only four (4) of these areas have planned or underway sewer expansion projects. For the remaining 99 areas, no funded, approved, or scheduled path to sewer connection exists, and without new planning and investment, there is no realistic pathway to connection before 2050.
- **Hazard and Feasibility Alignment:** When cost and hazard data are evaluated together, a small subset of areas emerges where favorable economics and high environmental urgency coincide. The tool identifies these areas, which represent the strongest targets for sewer investment statewide.
- **Additional Constraints on Sewer Expansion:** Sewer expansion faces several other significant headwinds identified through engagement with agencies and engineers. Disposal of treated effluent remains a critical unresolved constraint: new ocean outfalls are difficult to permit and injection well capacity is shrinking. County wastewater agencies are largely consumed by maintaining aging infrastructure and addressing expensive consent decree obligations, leaving limited funds for new expansion. Lower-cost alternative technologies such as low-pressure sewer are often presented as a more affordable alternative to gravity sewers; however, when operation and maintenance responsibilities fall partly on homeowners or future operators, the apparent public-sector savings may not reflect the total system cost over time.

Approach and Technical Results

During this study, all available sewer infrastructure and wastewater planning data were compiled statewide, including GIS layers, capital improvement program budgets, facility master plans, engineering studies, and service area records obtained through direct engagement with county wastewater agencies, private utilities, the Hawai‘i Department of Health, and federal partners. Data were integrated into a comprehensive database and used to delineate the 103 Potential Project Areas where cesspool density meets a minimum viability threshold for municipal-scale gravity sewer of 50 units per ⅛ mi^2. All underlying data, area-specific details, and findings are accessible at www.hawaiicesspooltool.org, through a newly developed interactive web application where users can explore the full statewide map and query Potential Project Area planning status, cost feasibility, and cesspool counts, as well as explore public and private wastewater infrastructure facilities and service areas for the state. Readers are encouraged to open the tool and explore their community or region of interest alongside this report.

Sewer Expansion Planning Status

No single agency maintains a statewide picture of where sewer expansion is planned or even under consideration. To address this, each Potential Project Area was assigned a planning status based on current county plans, capital improvement program records, and direct consultation with county wastewater agencies. Specifically these are:

- **Planned or Underway:** Formally approved, funded, or already under construction.
- **Under Evaluation:** Identified in plans or CIP budgets but not yet approved or funded.
- **No Active Plans:** No known study, or planning effort identifies the area for sewer expansion.

The results reveal a stark planning gap. Of the 103 Potential Project Areas, only 4 are Planned or Underway, covering ***less than 3 percent*** of the cesspools identified by this tool. Another 32 are Under Evaluation, but lack any known commitment to implementation. The remaining 67 areas, covering roughly 80% of project area cesspools, have no documented sewer planning of any kind (**Figure E1**).

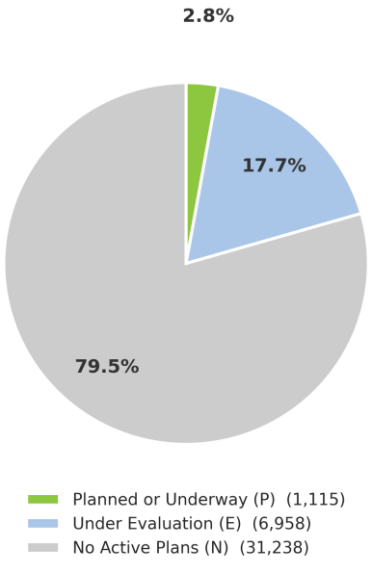


Figure E1. *Distribution of cesspools within the 103 Potential Project Areas by sewer expansion planning status.*

Evaluating Feasibility with a Planning-Level Cost Model

Cost is the most direct measure of whether sewer expansion is realistic for a given area. The more it costs to connect each cesspool, the harder it is to justify centralized sewer over on-site alternatives. To quantify this for each Potential Project Area, cost data were compiled from capital improvement program budgets, engineering studies, Clean Water State Revolving Fund materials, and county draft integrated wastewater management plans. These were used to derive unit costs for four infrastructure components: 1) collection pipe, 2) conveyance mains to the nearest treatment plant, 3) pump stations, and 4) adding wastewater treatment capacity. Statewide GIS analysis was then used to measure the area-specific physical inputs for each Potential Project Area, including road network length, distance to the nearest treatment facility, terrain slope, and substrate type. All necessary component costs for each project area were summed to calculate a modeled dollar cost to sewer each area. Because publishing raw dollar estimates for these modeled planning-level projections would imply a precision the model cannot support, the results are instead expressed as five relative cost feasibility tiers from 1 - $ (most feasible) to 5 - $$$$$ (least feasible), as shown in **Table E1.**"

Table E1. *Cost feasibility tiers applied to all 103 Potential Project Areas, expressed as relative cost ratings and approximate multiples of a typical onsite cesspool-to-septic conversion cost. The majority of areas fall in the two least feasible tiers.*

Rating	Relative Cost	Potential Project Areas	Percent of Total Project Areas	Approx. Multiple of Onsite Alternative[1]
(1) - $	Lowest relative cost	4	4%	Under 3x
(2) - $$	Low to moderate relative cost	7	7%	3x to 5x
(3) - $$$	Moderate relative cost	26	25%	5x to 8x
(4) - $$$$	High relative cost	44	43%	8x to 14x
(5) - $$$$$	Highest relative cost	22	21%	Over 14x

Identifying the Strongest Candidates for Sewer Investment

By combining the cost model results with the HCPT hazard framework, this analysis is able to answer a question no prior work has addressed statewide: **where do the most urgent cesspool problems overlap with the most economically viable sewer solutions?** The areas that offer the greatest return on public investment will be those where low construction cost, high cesspool hazard, and large scale of impact converge. Because a finer distinction than the five cost tiers is needed, dollar cost per connection is normalized by the statewide median to calculate the Median Relative Cost (MRC), a dimensionless index where 1.0 represents the typical project, 0.5 is half as expensive, and 2.0 is twice as expensive, allowing relative feasibility comparison across areas. **Figure E2** plots all 103 Potential Project Areas simultaneously by relative cost (MRC) and 'impact score', a composite of total cesspool count

weighted by HCPT priority, revealing where the most favorable opportunities for sewer investment lie.

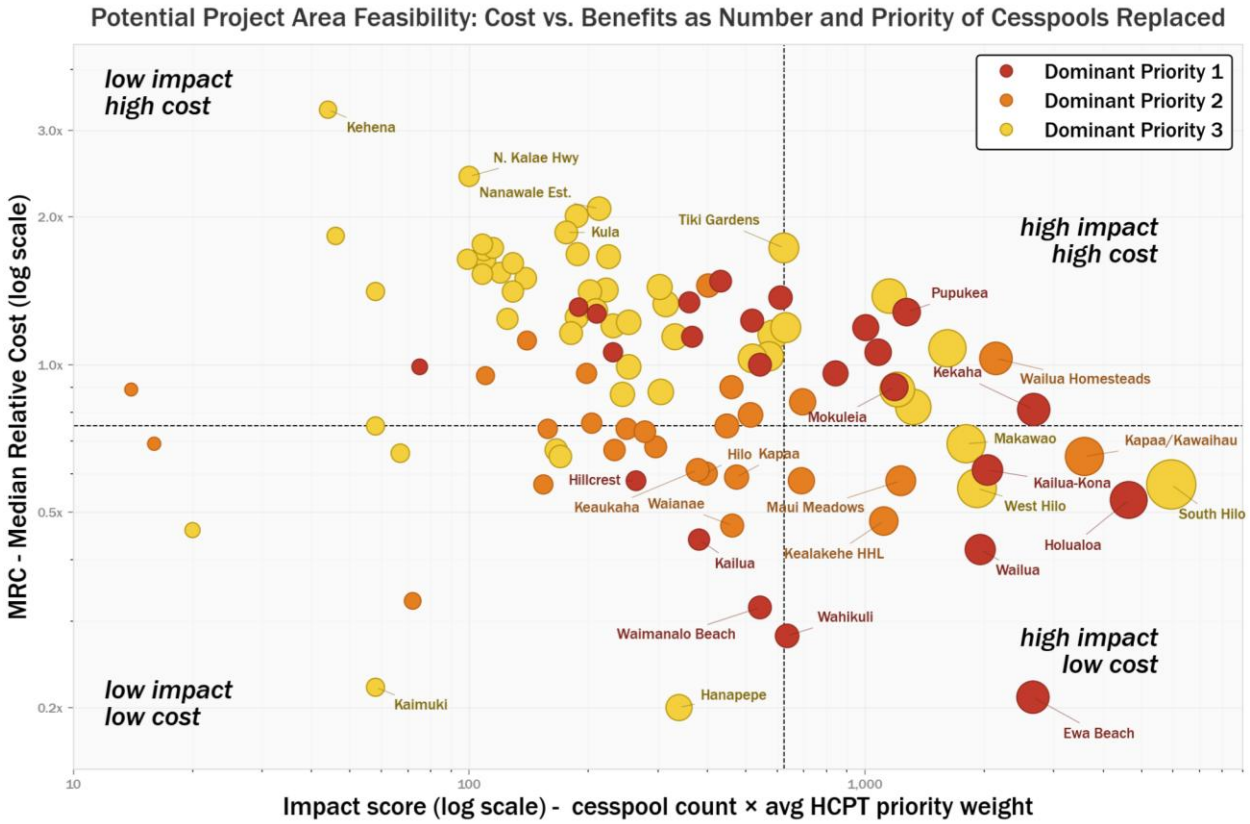


Figure E2. *Relationship between sewer construction cost and cesspool impact across all 103 Potential Project Areas. MRC (y-axis) is a dimensionless relative cost where 1.0 equals the statewide median. Impact Score (x-axis) combines total cesspool count with average HCPT priority weighting. Dashed lines mark statewide averages. Bubble size is proportional to cesspool count. Color indicates dominant HCPT priority class. Both axes are log scale. Lower-right quadrant areas offer the most favorable combination of low cost and high impact.*

Background

Extent and Impacts of Legacy Cesspools

Approximately 83,000 cesspools remain in active use across the State of Hawai‘i, collectively discharging an estimated 50 million gallons of untreated sewage per day into groundwater and surface waters (State of Hawai‘i Department of Health, 2022; Mezzacapo et al., 2020). In a state where 99 percent of drinking water is derived from groundwater (U.S. Geological Survey, 2023), and where permeable volcanic substrates allow rapid subsurface transport of contaminants to wells, streams, and nearshore waters (Izuka et al., 2018; Hondula et al., 2025), this volume of unmanaged wastewater constitutes one of Hawai‘i’s most pressing environmental infrastructure challenges. Cesspools provide no treatment of the wastewater they receive. They consist of porous, lined or completely unlined pits that allow raw sewage to leach directly into surrounding soils and groundwater (**Figure 1**). Cesspools are now prohibited in all 50 states as a permitted wastewater disposal method.

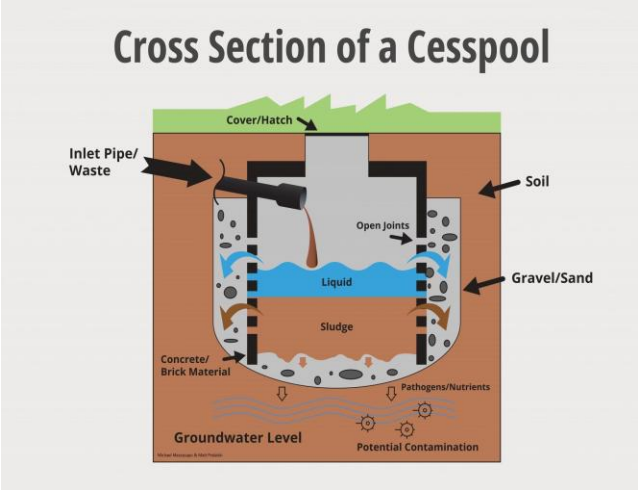


Figure 1. *Typical cesspool cross-section showing the discharge of untreated wastewater into surrounding soil and groundwater. Source: Wastewater Alternatives & Innovations.*

The approved onsite alternatives, septic systems and aerobic treatment units (ATUs), provide higher-level treatment before discharge to the subsurface. Unlike cesspools, septic systems include a tank for initial treatment and a soil absorption field where pathogens and nutrients are further attenuated. This treatment depends on adequate soil depth above the water table, and in Hawai‘i there are many places where soils are thin or nonexistent. Shallow water tables, thin soils, and highly permeable volcanic substrate reduce leach field performance to the point where, in the most vulnerable areas, septic systems offer little improvement over cesspools. Centralized sewer removes wastewater from the ground entirely and treats it at a dedicated facility, making it the most reliable solution where groundwater and coastal water protection is the goal.

Hawai‘i’s Hydrogeology Amplifies Cesspool Risks

The islands are composed primarily of young, highly permeable basaltic rock overlain in many areas by thin soils with high infiltration rates. Water tables are shallow in many coastal and low-lying areas, reducing the unsaturated soil zone available for natural attenuation of pollutants. Groundwater flow paths in volcanic aquifers are often short and fast, creating strong hydrologic connections between cesspool discharge points and receiving waters, including public drinking water wells, perennial streams, wetlands, and nearshore marine environments (Mezzacapo & Shuler, 2022) (**Figure 2**).



Figure 2. *Fluorescein dye emerges from shoreline springs along the Puakō shoreline following a residential dye tracer test. Dye tracer studies conducted by UH Hilo researchers showed hydrologic connections between onsite wastewater systems and nearshore waters (Wiegner et al., 2021)*

One practical consequence of Hawai‘i’s hydrogeology is that the performance assumptions underlying conventional onsite wastewater treatment systems do not hold equally across the state. The treatment potential of a soil absorption field depends on the presence of a sufficiently deep unsaturated soil zone between the leach field and the water table, a condition that is limited or absent in many coastal and low-lying areas of Hawai‘i. Where shallow water tables, thin soils, or highly permeable volcanic substrate bring cesspools and conventional septic leach fields into close proximity with the saturated zone, the distinction in treatment performance between these system types is substantially diminished.

The Puakō area of Hawai‘i Island illustrates this limitation. Wiegner et al. (2021) found that the thick soil layer needed for microbial breakdown below septic leach fields is largely absent; groundwater is close to the land surface, and the substrate is highly permeable fractured basalt rather than soil, limiting the breakdown and filtering of wastewater pollutants before they reach groundwater and the coast. A statewide University of Hawai‘i assessment reached a similar

conclusion at a broader scale, finding that wastewater impacts are concentrated where high cesspool density overlaps with shallow groundwater, permeable substrate, and direct land-to-sea groundwater flow paths.

The distribution of impacts is therefore not uniform. Cesspools in areas with shallow water tables, high soil permeability, proximity to drinking water sources, shoreline springs, streams, or sensitive nearshore ecosystems pose substantially greater risks than systems in areas with deeper water tables, thicker soils, or greater distance from sensitive receptors. Documented effects span multiple sectors: nutrient loading contributes to algal overgrowth on coral reefs, pathogen transport threatens recreational water quality, and contaminants of emerging concern have been detected in nearshore waters receiving wastewater-influenced groundwater. These spatial differences make a uniform replacement timeline impractical and a prioritization framework essential.

Environmental and Human Health Impacts of Cesspools

Where cesspool-derived contamination reaches aquifers and coastal waters, the consequences span human health, drinking water supply, and nearshore ecology. In communities with high cesspool density, private drinking water wells have tested positive for fecal indicator bacteria at rates that indicate widespread groundwater contamination. At Hawaiian Paradise Park on Hawai‘i Island, DOH sampling found 50 percent of tested private wells positive for fecal indicator bacteria (Whittier & El-Kadi, 2014). Nitrate concentrations in cesspool-impacted groundwater have been documented above the EPA maximum contaminant level of 10 mg/L nitrate-nitrogen, a threshold established specifically to protect bottle-fed infants under six months of age, for whom elevated nitrate intake causes methemoglobinemia by impairing the oxygen-carrying capacity of blood (Mezzacapo et al., 2020). Fecal indicator bacteria are similarly elevated in coastal and recreational waters adjacent to cesspool-dense areas, with monitoring at Puakō documenting exceedances of DOH water quality standards at shoreline stations fronting 81 percent of sampled homes (Wiegner et al., 2021; Miller-Pierce & Rhoads, 2019).

In nearshore waters, the nitrogen load delivered via submarine groundwater acts as a nutrient subsidy that drives excessive macroalgal growth, particularly of non-native species whose competitive responses to enriched conditions differ from those of native benthic communities (Amato et al., 2016; Prouty et al., 2017; Richards Donà et al., 2023). Dense algal canopy shades coral and suppresses calcification, contributing to documented shifts from coral-dominated to algae-dominated reef states in areas of high onsite system influence (Yoshioka et al., 2016; Amato et al., 2016). As algal biomass accumulates and decomposes, microbial respiration consumes dissolved oxygen, producing the hypoxic conditions that define coastal eutrophication and further reduce reef resilience (Mezzacapo et al., 2020; Mezzacapo & Shuler, 2022).

Legislative and Regulatory Response

The State of Hawai‘i has responded to the threats to human health and the environment posed by cesspools primarily through legislation and regulation at the state level (Table 1). Earlier updates to Hawai‘i Administrative Rules (HAR 11-62), administered by the Department of Health, prohibited new cesspool construction by requiring permitted wastewater systems to meet treatment and disposal standards that cesspools do not satisfy. Act 125 (2017) built on that foundation, mandating that all existing cesspools be upgraded, converted, or connected to approved wastewater systems by January 1, 2050, making Hawai‘i the first state in the nation to adopt a comprehensive replacement timeline for existing systems rather than merely prohibiting new ones (Mezzacapo & Lerner, 2019; State of Hawai‘i Department of Health, Environmental Management Division, 2017). Act 132 (Hawai‘i State Legislature, 2018) established the Cesspool Conversion Working Group (CCWG) within DOH to develop a long-range conversion plan and funded a statewide nearshore contamination study that provided critical baseline data on the extent of wastewater impacts to coastal waters. Working in parallel under the broader Act 132 research mandate, WRRC and Sea Grant developed a new GIS-based cesspool hazard prioritization tool, which became the Hawai‘i Cesspool Prioritization Tool (HCPT) (Mezzacapo & Shuler, 2022).

The State Department of Health sets standards and regulates wastewater systems, but counties are responsible for planning, constructing, and maintaining sewer infrastructure. The 2050 mandate was established at the state level; the infrastructure decisions that will determine whether centralized service reaches a given neighborhood are made at the county level. Yet counties are not, as of the time of publication of this report, required to develop wastewater management plans, nor are they required to identify which areas could feasibly connect to existing or planned treatment facilities. No single agency maintains a statewide picture of where sewer expansion is planned, where it is feasible but unplanned, and where it is unlikely to occur within the mandate timeframe.

Table 1. *Summary of legislative and regulatory actions that have shaped the state's response to cesspool pollution. Each milestone is a state or federal action; no significant corresponding county-level planning requirement or action appears in the record.*

Year	Action	Type	Key Provisions
2000	EPA UIC Rule	Federal regulation	Prohibited construction of new large-capacity cesspools nationwide; required closure of all existing large-capacity cesspools by April 5, 2005.
2015–2016	HAR 11-62 Amendment	State regulation	Prohibited construction of new individual cesspools statewide; did not require conversion of existing systems.
2017	Act 125, SLH 2017	State legislation	Mandated upgrade, conversion, or sewer connection of all existing cesspools by January 1, 2050; established exemption criteria; directed DOH to investigate cesspool scope and priority statewide.
2018	Act 132, SLH 2018	State legislation	Established the CCWG within DOH to develop a long-range statewide conversion plan; directed WRRC to conduct a statewide nearshore contamination study; authorized WRRC and Sea Grant to develop a GIS-based cesspool hazard prioritization tool, which became the HCPT.
2022	HCPT	Tool and report	Statewide GIS-based cesspool hazard prioritization tool released publicly at hawaiicesspooltool.org; census-tract level prioritization across fifteen risk factors.
2024	Act 217, SLH 2024	State legislation	Directed WRRC and Sea Grant to develop a centralized system expansion overlay to the HCPT; specified five feasibility factors—density of development, proximity to wastewater infrastructure, existing county plans, timing, and cost—plus any other factors deemed relevant; unencumbered appropriated funds lapse June 30, 2026.

The Hawai‘i Cesspool Prioritization Tool

The actions summarized in **Table 1** led to the development of the Hawai‘i Cesspool Prioritization Tool (HCPT), the State’s primary framework for identifying where cesspools pose the greatest relative hazard (**Figure 3**). The HCPT was developed by the University of Hawai‘i at Mānoa Water Resources Research Center and Sea Grant College Program for the Department of Health and the Cesspool Conversion Working Group. It provides a GIS-based assessment of cesspool hazard across Hawai‘i using fifteen environmental and human-use factors, including proximity to drinking water sources, depth to groundwater, soil characteristics, cesspool density, coastal recreation usage rates, and ecological sensitivity.

The HCPT combines all of the fifteen environmental and human health indicators into a single metric to assign a relative cesspool hazard, or priority level, at the census-tract scale. Its priority levels identify where cesspool impacts are most likely to affect drinking water, surface waters, nearshore ecosystems, and other sensitive receptors. The tool does not assign replacement technologies or determine whether sewer connection is feasible for a given property or community. The HCPT classifies areas into three priority levels, providing the State with the preeminent framework for identifying where cesspool impacts are most acute and where intervention is most urgent.



Figure 3. *The Hawai‘i Cesspool Prioritization Tool interactive web map, showing statewide cesspool hazard priority tiers. Accessible at: www.hawaiicesspooltool.org*

The Planning Gap

The HCPT was designed to answer where cesspools pose the greatest hazard. It was not designed to answer the equally important questions of how those cesspools should be replaced, by whom, at what cost, or on what timeline. That limitation is not a flaw in the tool; it reflects a structural feature of wastewater governance in Hawai‘i. The Department of Health sets standards and regulates wastewater systems, but counties are responsible for planning, constructing, and maintaining sewer infrastructure. The 2050 mandate was established at the state level; the infrastructure decisions that will determine whether centralized service reaches a given neighborhood are made at the county level. Yet counties are not, as of the time of publication of this report, required to develop wastewater management plans, nor are they required to identify which areas could feasibly connect to existing or planned treatment facilities. No single agency maintains a statewide picture of where sewer expansion is planned, where it is feasible but unplanned, and where it is unlikely to occur within the mandate timeframe.

The HCPT represents meaningful progress, establishing a statewide basis for identifying which cesspools pose the greatest hazard and where intervention is most urgent. But knowing where to act is not the same as knowing how. For individual cesspool owners, the absence of a coordinated state and county strategy for replacement pathways leaves a consequential question unanswered: whether to invest in onsite conversion now, or wait for centralized wastewater service that may or may not materialize. A property owner in a high-priority area may face costs of tens of thousands of dollars either way, with no reliable information about whether sewer service is planned for their area, physically feasible, or likely within any defined timeframe. Those who convert independently may find that sewer service arrives shortly afterward, rendering their investment partially redundant. Those who wait may find that no centralized service is forthcoming and that they have delayed compliance or development of their property unnecessarily.

Counties face a parallel challenge. Sewer expansion is capital-intensive, and counties must simultaneously maintain and operate existing infrastructure, fund required upgrades, and meet legal compliance obligations, including consent decree requirements that have directed substantial capital toward existing system repair and rehabilitation in recent years. Expansion into unserved areas must compete for resources within that constrained fiscal environment. Without a statewide analytical framework that links cesspool hazard priorities to the feasibility and cost of centralized service expansion, counties lack a systematic basis for identifying where sewer extensions would serve the greatest public benefit per dollar invested. The State, in turn, lacks the information needed to coordinate across counties, target financial assistance programs, and evaluate progress toward the 2050 mandate at a level more specific than the broad priority tiers provided by the HCPT.

Closing the Gap

This report and the accompanying tool represent a meaningful step toward closing that gap, providing property owners, counties, and the Legislature with a clearer basis for determining where centralized service is a realistic option and where onsite replacement is the more likely near-term path. It is also the first work in the legislative and regulatory timeline to begin bridging the divide between state-level mandate and county-level planning, laying groundwork for the coordinated effort that meeting the 2050 deadline will ultimately require.

Methodology

Data Collection and Stakeholder Engagement

To support defensible statewide comparisons of sewer expansion feasibility, all available sewer infrastructure and wastewater planning data were compiled from every accessible source, including direct consultation with county wastewater departments. Sewer mains, laterals, and service area boundaries covering both public and private wastewater systems were compiled from the Public Utilities Commission (PUC) database, Hawai‘i Statewide GIS Program, and directly from private wastewater companies as well as county environmental departments. A systematic literature review of online planning documents relating to wastewater projects across the state was conducted, including county Capital Improvement Program (CIP) budgets, wastewater facility master plans, engineering planning studies, environmental assessments, and service area definitions for private utilities as filed with the PUC. Relevant documents were publicly archived, and where appropriate, data from the literature review was digitized and integrated into geospatial and tabular databases, all made available in the publicly accessible online repository at the following web link: (https://github.com/cshuler/WW_Overlay_2024).

Data collection was supplemented by direct stakeholder engagement throughout the project. The project team met in person and virtually with all four county wastewater agencies, some private wastewater companies, and the Hawai‘i Department of Health on multiple occasions beginning in February 2025, maintaining ongoing email coordination between meetings. Counties provided CIP data, GIS shapefiles, project timelines, and iterative review of planning status classifications. The Hawai‘i Department of Health provided datasets and methodological consultation. The U.S. EPA Office of Wastewater Management and its contractors Eastern Research Group and Carollo Engineers, through the Closing America’s Wastewater Access Gap Initiative, collaborated with the authors on state cesspool conversion goals and methodology to align the density-based, cesspool-replacement planning approach with their cesspool conversion evaluations reports (Carollo Engineers, Inc., 2026).

Identification of Potential Project Areas

A consistent geographic unit was needed to enable systematic, statewide comparison across areas with vastly different scales, costs, and conditions. For this analysis, discrete polygon areas were delineated based on cesspool density and defined as **Potential Project Areas.** These areas are locations where cesspool density meets a minimum threshold for centralized gravity sewer to be economically viable. The threshold adopted here was 50 or more cesspools within one-eighth of a square mile, established through coordination with county wastewater departments and their consultants Carollo Engineers. This threshold represents the minimum density at which per-connection sewer costs are reasonable, and is adopted directly from the counties' own planning standard, maintaining consistency with the county integrated wastewater management plans currently being developed by Carollo Engineers.

Area boundaries were drawn through a four-step process. Automated density analysis was conducted on the 2022 HCPT cesspool inventory (Mezzacapo & Shuler, 2022) to identify candidate areas meeting the 50 per eighth square mile threshold. Boundaries were then manually drawn as contiguous polygons refining the automated outputs. A 150-meter proximity buffer was applied to identify adjacent areas suitable for consolidation into larger service areas, and combined boundaries were redrawn based on geographic constraints including roads, topographic breaks, and natural boundaries (**Figure 4**). A small number of additional areas that did not meet the density threshold but were identified by counties as areas of interest were also included as potential project areas. The process yielded 103 Potential Project Areas statewide.

It bears noting that Potential Project Areas are defined specifically as candidates for conventional gravity sewer expansion. All four county wastewater agencies operate gravity systems as their standard, and county staff consistently indicated that low-pressure and grinder pump alternatives are not preferred as they shift long-term maintenance costs to homeowners and require equipment and operator training the counties may not have. Areas unsuitable for gravity sewer are outside this tool's intended scope and better evaluated under a separate framework.

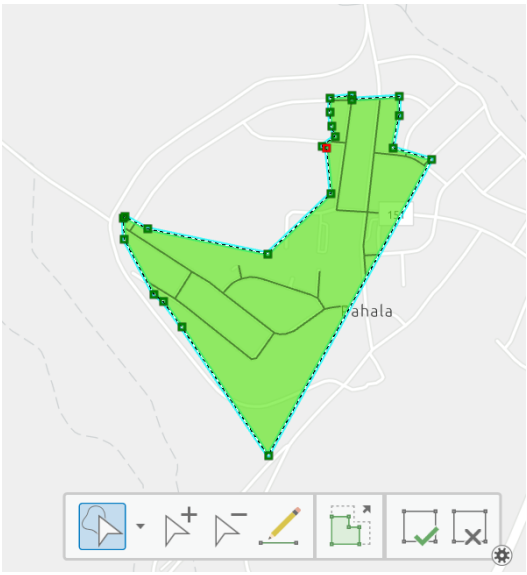


Figure 4. *Example of the potential project area boundary refinement process during delineation.*

Assigning Sewer Expansion Planning Status

Each Potential Project Area was assigned a planning status describing the documented level of county, utility, or community-based, area-specific activity associated with centralized sewer expansion (**Figure 5**). Planning status was assigned from available county plans, capital improvement program records, published planning studies, utility service area information, and direct consultation with county wastewater agencies. These classifications are intended to support planning-level comparison across areas; they do not constitute engineering determinations, funding commitments, permit approvals, or final recommendations for a specific replacement pathway. Planning status reflects conditions at the time of data collection and will be updated periodically as the tool is maintained. For areas served by private utilities, planning status was based on PUC filings and direct outreach where possible, but may be less current than for county-operated systems.

The planning status categories defined for each project area are:

Planned or Underway (P). Areas where a sewer expansion project has been formally approved, funded, scheduled, or is already under construction or active implementation. This category also includes projects confirmed by the responsible wastewater utility provider as expected to be built before 2050.

Under Evaluation (E). Areas where centralized sewer expansion has been identified in planning documents, studies, published capital improvement program materials, or county or utility discussions, but has not been formally approved, funded, or scheduled for implementation. These areas remain uncertain and require additional engineering, funding, policy, or community decision-making before they can be treated as likely projects.

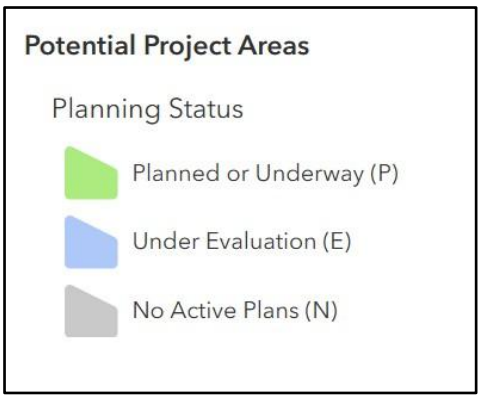


Figure 5. *Planning status categories as displayed in the web tool.*

No Active Plans (N). Areas where no known sewer expansion project, study, proposal, or planning effort currently identifies the area for centralized sewer infrastructure. These areas contain cesspools at sufficient density to make centralized wastewater service a potential future option, but no documented planning activity was found at the time of analysis.

Modeling Feasibility Through Cost Analysis

Cost is the primary feasibility metric used in this analysis, expressed on a per-cesspool basis to allow direct comparison across Potential Project Areas of vastly different sizes and configurations. For each area, total planning-level capital cost is estimated as the sum of four infrastructure components: (1) collection mains, (2) conveyance mains, (3) pump stations, and (4) wastewater treatment capacity, in the form of building new or upgrading existing wastewater treatment plants (WWTPs). Each component is calculated by applying a unit cost rate derived from documented Hawai‘i wastewater project data to a physical quantity measured or estimated for each specific Potential Project Area through GIS analysis and stakeholder consultation. The data sources, input quantities, unit costs, and calculation method for each component are summarized in **Table 2.**

Table 2. *Cost model structure: data sources, area-specific input quantities, unit costs, and calculation method for each of the four infrastructure cost components used to estimate total planning cost per Potential Project Area.*

Cost Component	GIS/Data Input	Input Quantity	Unit Cost	Cost Calculated As
Collection mains	State road network	Total road network length within area boundary (linear feet)	$1,902/LF	Length × unit cost × complicating factors
Conveyance mains	Road network + WWTP locations	Shortest road distance to nearest WWTP or sewer main (linear feet)	$3,302/LF	Distance × unit cost × complicating factors
Pump stations	Road network + terrain elevation data	Count of accumulation points from slope-based network analysis	$750k neighborhood / $6.5M regional	Count × unit cost
Treatment capacity	County wastewater departments, PUC filings	Cesspool count + capacity status	$147k-$19k/ connection (tiered by size)	Connections × tier rate (or $0 if capacity available)

Because the underlying cost model carries significant planning-level uncertainty, results are presented as relative tiers rather than absolute dollar figures. Five tiers are used, ranging from lowest to highest relative cost. For reference, in **Table 3** they are expressed as approximate multiples of a typical onsite cesspool-to-septic conversion cost to provide an intuitive sense of scale. Full technical documentation of the cost model, including unit cost derivation, GIS network analysis methodology, complicating factor adjustments, and model limitations, is provided in **Appendix A.**

Table 3. *Relative cost feasibility tiers applied to all Potential Project Areas,*

Rating	Relative Cost	Approx. Multiple of Onsite Alternative[1]
(1) - $	Lowest relative cost	Under 3x
(2) - $$	Low to moderate relative cost	3x to 5x
(3) - $$$	Moderate relative cost	5x to 8x
(4) - $$$$	High relative cost	8x to 14x
(5) - $$$$$	Highest relative cost	Over 14x

[1] *Approximate multiple of a typical onsite cesspool-to-septic conversion (Ishida, 2023; Azambuja, 2024). Provided for scale reference only.*

Results and Key Findings

Scope and Coverage of the Act 217 Sewer Expansion Overlay Tool

Act 217 directed the University of Hawai‘i to identify where centralized sewer expansion is most feasible as a cesspool replacement strategy. To accomplish this, the study delineated 103 Potential Project Areas statewide. These are discrete neighborhoods where cesspool density meets the minimum threshold (50 per ⅛ mi²) for municipal gravity sewer to be economically viable. Together these areas encompass approximately 39,300 cesspools, representing nearly half of the statewide inventory. The 103 Potential Project Areas and all underlying data are accessible through a dedicated interactive web tool at: https://seagrant.soest.hawaii.edu/cesspool-tool-wastewater-overlay.
The Act 217 Sewer Expansion Overlay Tool page is a sub-page of and hosted on the Hawai‘i Cesspool Prioritization Tool website www.hawaiicesspooltool.org, and is where users can explore the full statewide map, query individual areas by address or location, and view area-specific details for each Potential Project Area. Readers are encouraged to open the tool and explore their community or region of interest alongside this report.

The tool displays an interactive map of all Potential Project Areas color-coded by planning status and labeled with cost feasibility tier. It also includes public and private wastewater service areas, county and major private treatment plant locations, and proximity to existing sewer infrastructure **(Figure 6**). Clicking any project area opens a summary report showing planning status and details, cost feasibility tier, cesspool counts by HCPT priority class, total effluent discharge, nitrogen load, infrastructure requirements including collection pipe length, conveyance distance, and pump station counts, and links to relevant planning documents where available. County sewer mains and individual cesspool locations can be toggled on as reference layers. Users can export area summary reports and map views as PDF or image files directly from the tool. Additionally, all underlying geospatial and tabular data, code, and documentation are publicly available at https://github.com/cshuler/WW_Overlay_2024.

The tool’s database covers a substantial share of the state's highest-priority cesspools. Of the 14,541 Priority 1 cesspools statewide, 8,816 (61%) fall within the 103 Potential Project Areas. Priority 2 coverage is similar at 60%. Priority 3 coverage is lower at 41%, as lower-hazard cesspools tend to be more dispersed and less likely to meet the density threshold (**Table 4**). This reflects a consistent settlement pattern: older coastal communities built before modern wastewater infrastructure tend to concentrate people, environmental sensitivity, and cesspool density in the same places. The remaining 44,070 cesspools outside the 103 Potential Project Areas, including 5,685 Priority 1 cesspools, are generally too dispersed for municipal gravity sewer and will likely require on-site solutions.

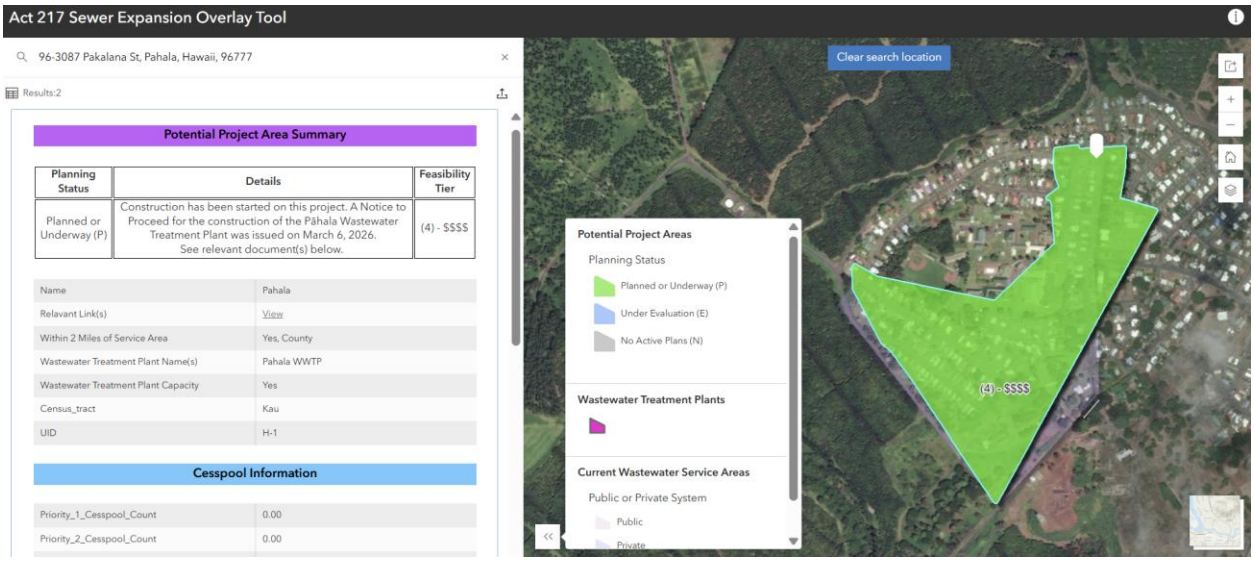


Figure 6. *The Act 217 Centralized Wastewater Overlay Tool interface. The map displays Potential Project Areas color-coded by planning status alongside wastewater treatment plant and service area layers. The left panel shows the area summary report generated when a user selects a Potential Project Area.*

Table 4. *Hawai‘i Cesspool Prioritization Tool priority rankings statewide and those falling within the 103 Potential Project Areas, representing high-density cesspool areas that meet the minimum threshold for municipal sewer consideration.*

HCPT Priority Rank	Statewide Cesspool Count	In Project Areas	% Within Project Areas
Priority 1 (High)	14,541	8,816	60.9%
Priority 2 (Medium)	12,367	7,431	60.1%
Priority 3 (Low)	55,797	23,064	41.3%
Unranked	716	--	--
Total	**83,421**	**39,311**	**47.1%**

Key Finding 1: Active Sewer Projects Cover Less than 4 Percent of Potential Project Areas

Each potential project area was assigned one of three planning statuses based on current published plans, CIP budgets, and information collected directly from county wastewater departments (**Table 5**). Planned or Underway (P) means a sewer project has been formally approved or scheduled, including anything under construction or expected to be completed before 2050. Under Evaluation (E) means a project appears in planning documents, studies, or CIP budgets but has not been funded, approved, or scheduled. No Active Plans (N) means no known project or study identifies the area for sewer, even though its cesspool density indicates it could be a candidate for centralized sewer.

The breakdown shows a significant shortfall in active planning; of the 103 project areas, only 4 are Planned or Underway, accounting for 1,115 cesspools. This represents a mere 1.4 percent of all cesspools statewide and 2.9 percent of those within the project areas (**Figure 7**). Another 32 areas are classified as Under Evaluation, covering 6,958 cesspools, but the viability of these projects remains uncertain. The remaining 67 areas, which cover 31,238 cesspools or roughly 80 percent of all cesspools identified by this tool, have no active sewer plans at all. Put simply, four out of five cesspools that may be suited to sewer have no project plans of any kind behind them.

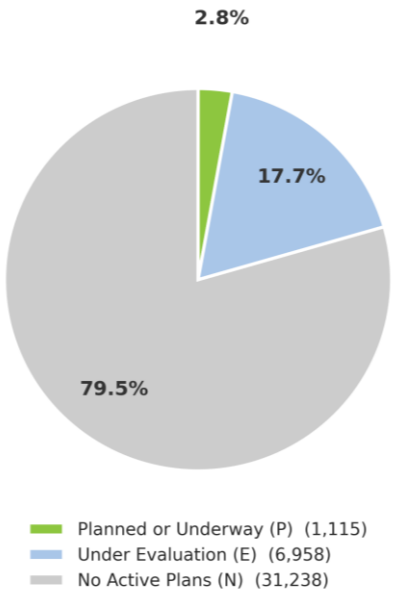


Figure 7. *Distribution of cesspools within the 103 Potential Project Areas by sewer expansion planning status.*

Table 5*. Sewer expansion planning status across the 103 Potential Project Areas, showing cesspools by status as a share of both the project area total and the statewide cesspool inventory.*

Planning Status	Project Areas	Cesspools	% of Project Area Total	% of Statewide Total
Planned or Underway (P)	4	1,115	2.8%	1.3%
Under Evaluation (E)	32	6,958	17.7%	8.3%
No Active Plans (N)	67	31,238	79.5%	37.4%
Total	**103**	**39,311**	**100.0%**	**47.1%**

The complexity and typically long timelines of large wastewater projects make this gap even more urgent than it first appears. Moving a sewer project from initial planning through design, permitting, funding, and construction likely takes 10 to 15 years or more, and public opposition can stall projects further. Working back from the 2050 deadline, any area not already in active planning by roughly 2035 is unlikely to be connected in time. With 67 potential project areas holding 80 percent of potential project area cesspools in the No Active Plans category, the practical window to even begin most of these projects is closing within the next decade.

Key Finding 2: Feasibility Is the Exception: Most Project Areas Fall in the Highest Cost Tiers

Each Potential Project Area was also assigned a cost feasibility tier from 1 - $ (most feasible) to 5 - $$$$$ (least feasible), based on a planning-level cost model developed for this analysis using Hawai‘i-specific cost data. Cost per connection is used as a proxy for economic feasibility: the more it costs to connect each home, the less practical sewer becomes relative to on-site alternatives. The cost analysis paints a consistent picture: connecting cesspools to sewer is expensive across the board. There are no cheap solutions, only less expensive ones. Of the 103 project areas, only 11 fall in the two most feasible tiers. Another 26 areas fall in Tier 3, with the majority, 66 project areas, sitting in Tiers 4 and 5, the least feasible (**Figure 8**).

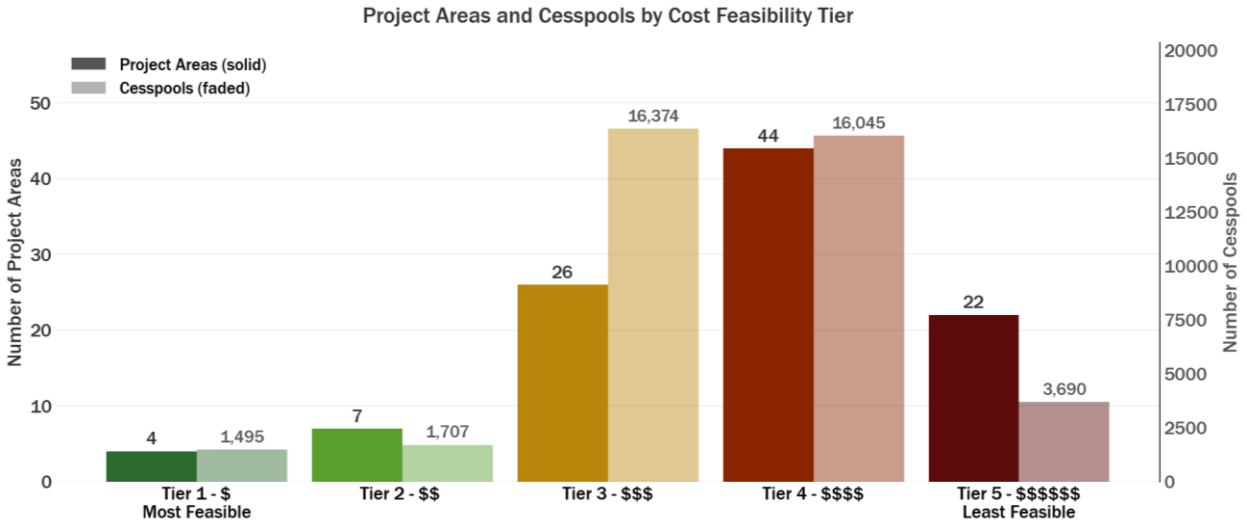


Figure 8*. Number of Potential Project Areas and total cesspools within each cost feasibility tier, where Tier 1 represents the most feasible and Tier 5 the least feasible based on estimated cost per connection. Solid bars reference the left axis; faded bars reference the right axis.*

The cost model was developed using data-informed cost values for four infrastructure components for each area: collection piping, conveyance piping to the nearest treatment plant, treatment plant construction or upgrades, and neighborhood and regional pump stations, adjusted for steep terrain and hard rock substrate that raise excavation costs. The project area total is divided by the number of connections to yield a per-connection cost, then sorted into one of five tiers. Only the relative cost feasibility tiers are reported, not underlying dollar estimates. The cost model is a planning-level tool, not an engineering estimate, and has not been validated against actual project bids. Tiers communicate relative feasibility across areas without implying a precision the model was not designed to provide. Where a more refined assessment of the variability is needed, cost is also expressed as the Median Relative Cost (MRC), a dimensionless value where 1.0 represents the cost of the median project. An area with an MRC of 2.0 is roughly twice as expensive per connection as the typical project; an area at 0.5 is half as expensive.

Even without analyzing actual costs, the cost model yields insights about the relative feasibility of sewering the different potential project areas (**Figure 9**). The least costly area, Hanapēpē on Kauaʻi, is roughly 17 times less costly per connection than the most expensive, Kehena on Hawaiʻi Island. The spread demonstrates how sewer feasibility varies enormously by location. For most project areas the per-connection cost is probably high enough that public investment is difficult to justify on economics alone.

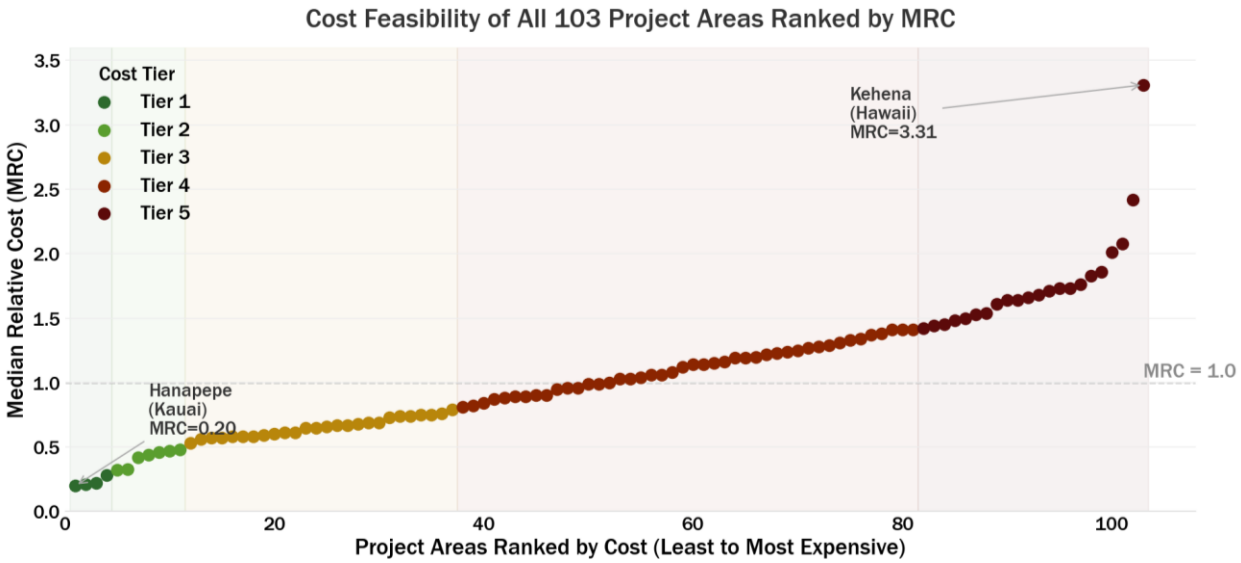


Figure 9. *All 103 Potential Project Areas ranked by Median Relative Cost (MRC), where MRC 1.0 represents the median project cost. Colors indicate cost feasibility tier, with Tier 1 (dark green) representing the most feasible and Tier 5 (dark red) the least feasible. Shaded regions denote tier boundaries.*

Key Finding 3: Cost and Hazard Data Identifies the Highest-Return Opportunities for Sewer Expansion

Cesspool conversion decisions involve tradeoffs across multiple dimensions simultaneously. Effective prioritization requires weighing cost, environmental urgency, and the scale of impacts together. This cost model and the HCPT hazard framework were developed independently, but together they are able to answer the core policy question: **where do the most urgent cesspool problems overlap with the most economically viable sewer solutions?**

Three variables determine which areas offer the greatest return on investment:

- **Cost**: Expressed as the Median Relative Cost (MRC), the estimated cost to construct each project area, on a relative scale.
- **HCPT Priority Level**: the severity of water quality and public health risk, as classified by the Hawai‘i Cesspool Prioritization Tool
- **Scale**: total cesspool count, which directly scales to the number of homeowners served and the volume of nitrogen removed from the environment. In the figure below, HCPT Priority Level and total cesspool count are combined into a single factor: “Impact Score”.

Figure 10 shows all 103 Potential Project Areas plotted against these three variables. The y-axis shows cost as the MRC, where lower values indicate more economically favorable projects. The x-axis shows “Impact Score”, a composite of total cesspool count weighted by HCPT priority, where Priority 1 cesspools carry 3x the weight, and Priority 2 carries 2x the weight of Priority 3. Bubble size is proportional to total cesspool count. The dashed lines mark the statewide averages on each axis. Areas in the lower-right quadrant combine above-median impact with below-median cost, indicating where the most favorable combination for sewer investment lies.

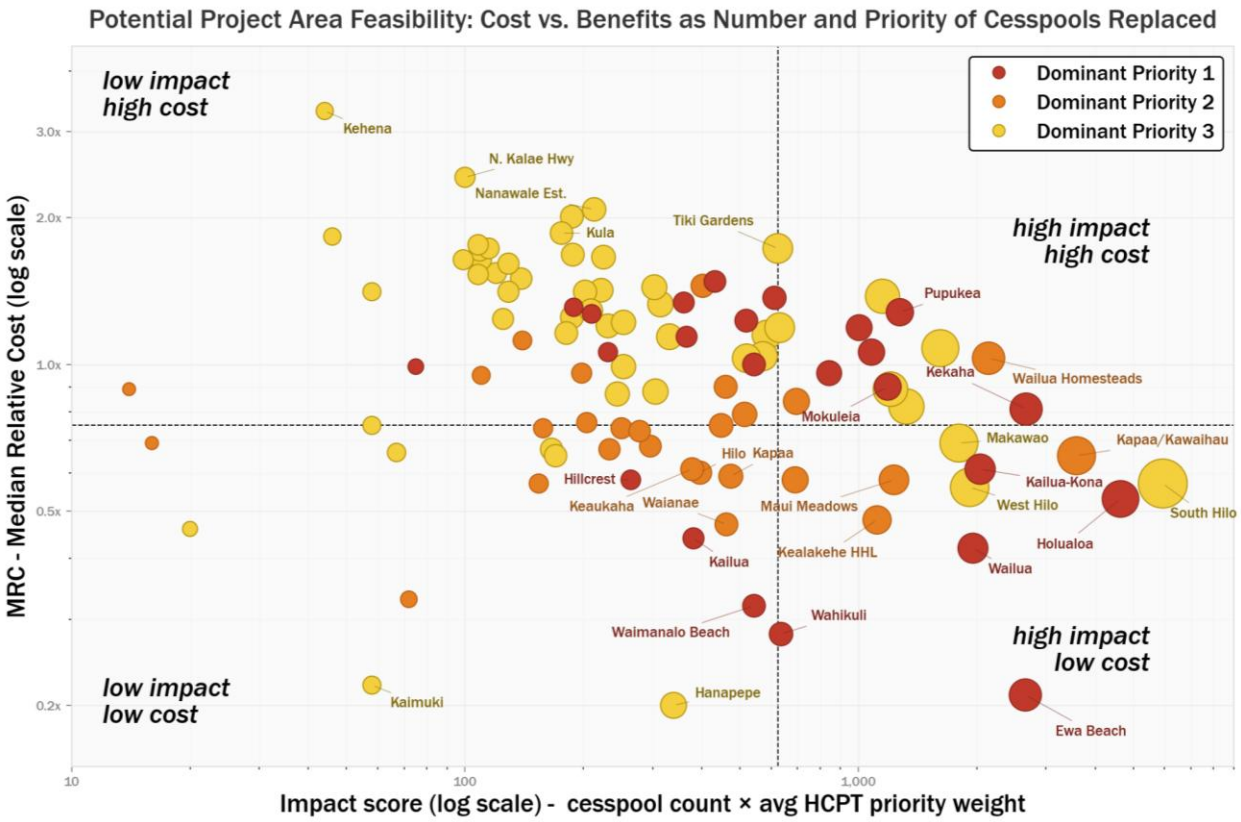


Figure 10. *Relationship between sewer construction cost and cesspool impact across all 103 Potential Project Areas. MRC (y-axis) is a dimensionless relative cost where 1.0 equals the statewide median. Impact Score (x-axis) combines total cesspool count with average HCPT priority weighting. Dashed lines mark statewide averages. Bubble size is proportional to cesspool count. Color indicates dominant HCPT priority class. Both axes are log scale. Lower-right quadrant areas offer the most favorable combination of low cost and high impact.*

The most actionable opportunities for sewer system expansion lie in the lower-right quadrant. These potential project areas combine high impact scores with low construction cost, meaning investments there would convert a large number of high-priority cesspools at a relatively favorable cost. ʻEwa Beach (O-11, Oʻahu), Wahikuli (M-4, Maui), and Wailua (K-5, Kauaʻi) demonstrate these economics. South Hilo (H-14, Hawaiʻi Island) is notably in this quadrant as it converts over 5,200 cesspools, albeit at a higher cost and at lower HCPT priority.

The upper-right quadrant contains high-impact areas where costs exceed the statewide median. Wailua Homesteads (K-7, Kauaʻi), Kalaoa (H-26, Hawaiʻi Island), and Pūpūkea (O-20, Oʻahu) fall here. These are not poor candidates on environmental grounds, but would likely require stronger justification or alternative funding to advance.

The lower-left quadrant contains areas that are relatively affordable but score below the mean on impact. Any cesspool converted is a net environmental benefit, but these projects do not

prioritize the most severe or widespread hazards first. Kaimukī (O-12, Oʻahu) and Hanapēpē (K-21, Kauaʻi) are representative examples, small areas with moderate cost profiles but limited cesspool counts and lower HCPT priority.

The upper-left quadrant represents the least favorable combination: below-median impact and above-median cost. Many of these are small, dispersed communities in difficult terrain. North Kalāe Hwy (Mo-2, Molokaʻi) and Kehena (H-21, Hawaiʻi Island) are the extreme examples, with very high MRCs and limited scale. These areas are better served by individual or small-scale decentralized solutions.

Table 6 identifies a selection of areas from or near the lower-right quadrant, distributed across the four main islands. These are not formal recommendations, they simply illustrate where the data point to the most favorable intersection of sewer expansion economics and cesspool hazard reduction on paper. Additionally, it should be kept in mind that this analysis is a statewide screening framework built on the best data available to the authors at this time. It is not an engineering study, and many of the key assumptions in the dataset are subject to revision as county partners review and update the data. In particular, WWTP capacity remains uncertain for several areas; a county correction to capacity status could significantly shift cost feasibility tier assignments. The areas identified here should be understood as candidates for further investigation and targets for planning, not as confirmed project priorities.

Table 6. *Selection of Potential Project Areas with favorable combinations of cost and impact score, by island.*

Area	Island	UID	Planning Status	% Priority-1	Total Cesspools	MRC	Cost Tier
ʻEwa Beach	O'ahu	O-11	P	100%	886	0.21	$
Waimānalo Beach	O'ahu	O-1	E	100%	181	0.32	$$
Kailua	O'ahu	O-2	E	100%	127	0.44	$$
Wahikuli	Maui	M-4	E	100%	252	0.28	$
Hanapēpē	Kaua'i	K-21	N	0%	339	0.20	$
Wailua	Kaua'i	K-5	N	100%	652	0.42	$$
Kealakehe HHL	Hawai'i	H-5	N	22%	503	0.48	$$
Holualoa	Hawai'i	H-17	N	100%	1549	0.53	$$$

Key Finding 4: Sewer Feasibility Varies Significantly Across Islands

The 103 Potential Project Areas are not distributed evenly across the state. They reflect the settlement history of each island, the location of existing sewer infrastructure, and the degree to which treatment and disposal capacity can be extended to nearby unsewered communities. The result is not a single statewide sewer problem, but a set of island-specific feasibility problems. **Table 7** summarizes the distribution of project areas and cesspools across islands and cost feasibility tiers, along with the median relative project cost for each island.

Table 7. *Distribution of Potential Project Areas and cesspools by island and cost feasibility tier*

Island	Total Areas	Number of Potential Project Areas In:					Total Cesspools	Median MRC
		Tier 1	Tier 2	Tier 3	Tier 4	Tier 5		
Hawai'i Island	42	-	1	10	16	15	20,003	1.29
Maui	16	1	1	6	4	4	5,103	0.82
Moloka'i	2	-	-	-	1	1	381	1.69
O'ahu	22	2	4	4	11	1	4,425	0.9
Kaua'i	21	1	1	6	12	1	9,399	0.84
Total	103	4	7	26	44	22	39,311	1

As would be expected, cost feasibility correlates with urbanization. Islands with denser development and more existing sewer infrastructure consistently show lower median MRCs. Oʻahu and Maui carry median MRCs below 0.9, while Kauaʻi sits slightly below the statewide median and Hawaiʻi Island and Molokaʻi sit well above it. On every island, more project areas fall in Tier 4 than any other tier. Expensive sewer projects are the norm statewide.

Three patterns emerge across the islands. First, planning activity does not track need. Kauaʻi's near-complete absence from the planning pipeline and Hawaiʻi Island's thin coverage relative to its scale suggest that where projects advance is driven more by county capacity and local initiative than by the statewide distribution of the problem. Second, scale and feasibility are inversely related where it matters most: Hawaiʻi Island and Kauaʻi carry the largest cesspool burdens but neither has meaningful planning momentum, while Oʻahu, the most feasible island, is also the only one with projects substantially under evaluation. Third, existing infrastructure is the strongest predictor of feasibility. Islands and areas near established sewer systems and treatment capacity consistently show lower costs, while remote communities on Molokaʻi and rural Hawaiʻi Island face structural cost disadvantages that geography and development history have locked in.

Key Finding 5: Sewer Cost Is Largely Predetermined by How Communities Were Built

When comparing project costs by individual components, three main patterns emerge. (1) collection pipe length, which is controlled largely by neighborhood density, is the single largest cost component. (2) access to and distance from existing WWTPs are major cost drivers. (3) because all costs are additive, the least feasible project areas are those where multiple cost drivers overlap. Across all 103 project areas, collection pipe accounts for 56% of total modeled cost statewide. Treatment plant construction or upgrades account for 20%, pump stations account for 12%, and conveyance to the nearest plant accounts for 12% (**Figure 11**).

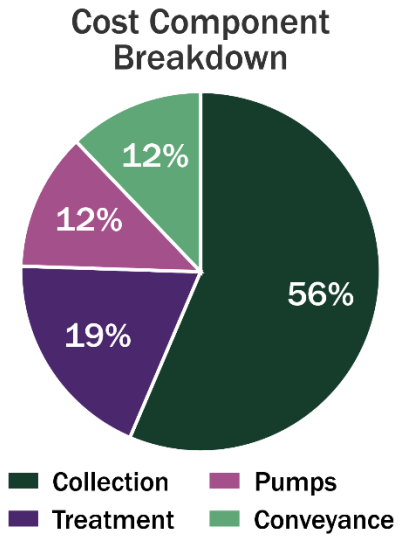


Figure 11. *Planning-level cost component breakdown across all 103 Potential Project Areas. Collection pipe represents the cost of gravity sewer mains within the project area; conveyance represents the cost of connecting the project area to the nearest wastewater treatment plant.*

This distribution reflects a defining feature of sewer retrofits, in that most of the cost stems from working underground. Unlike new development, where sewer is installed before roads are paved and homes are built, retrofit projects require excavating existing streets, managing traffic, working around buried utilities, and connecting individual homes to a central line. The pipe is expensive not just because of the material itself, but because of everything required to install it in an already-built environment. This makes the length of pipe, and thus neighborhood density one of the clearest drivers of cost. Spread out neighborhoods require more pipe to reach each cesspool, which raises cost before conveyance, pumping, or treatment needs are added. In this way, sewer feasibility is somewhat shaped by development decisions made decades ago, including lot size, road layout, and distance from existing infrastructure. This trend is clear when comparing collection and conveyance pipe requirements against total costs, as the MRC. Collection pipe reflects the internal geometry of each project area, including lot size, road length, and neighborhood density. Conveyance pipe reflects the area's distance from existing wastewater infrastructure. Together, the two measures show how both local development patterns and infrastructure access affect sewer feasibility. In **Figure 12** the most feasible projects cluster at the low end of pipe per connection, and the least feasible spread out to the right.

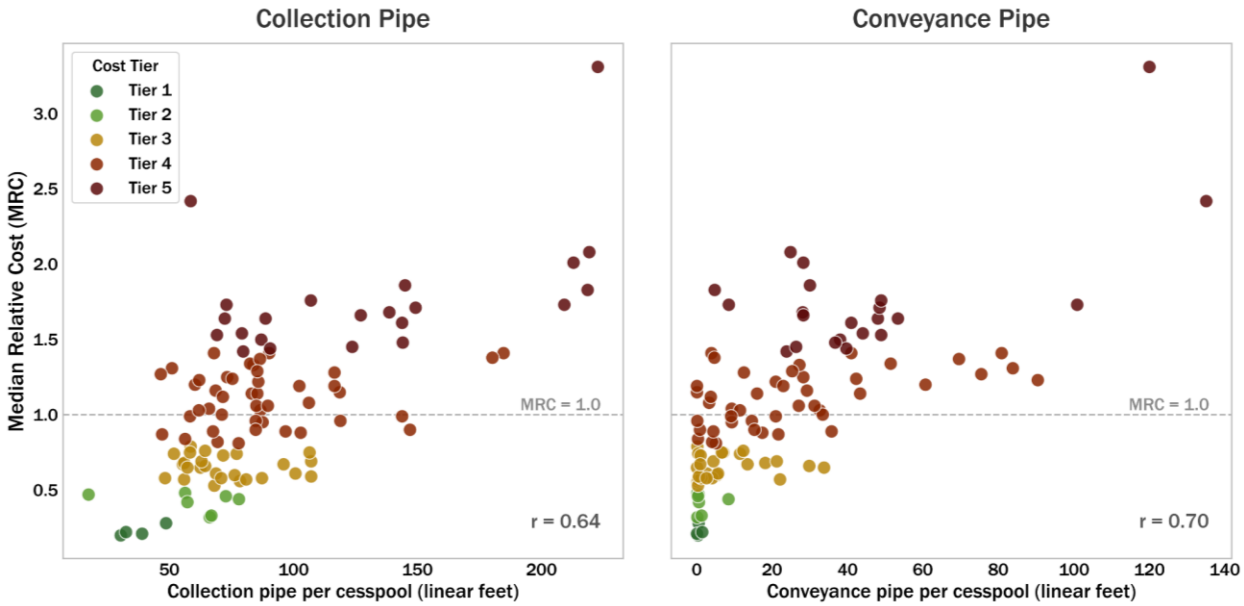


Figure 12. *Collection and conveyance pipe per cesspool versus Median Relative Cost (MRC) for all 103 Potential Project Areas, color-coded by cost feasibility tier. Collection pipe reflects the amount of infrastructure needed within each project area, while conveyance pipe reflects the distance needed to reach existing wastewater infrastructure. Areas requiring more pipe per cesspool generally show higher relative costs.*

The mix of cost components shifts as projects become more expensive. In the most feasible tiers, collection pipe accounts for 92% of total project cost. These projects are generally less expensive because they are both close to infrastructure and need little or no treatment work as well as being situated in more densely platted neighborhoods. As projects move into Tiers 4 and 5, the collection share drops to around 50%, not because pipe costs less in those areas, but because treatment and conveyance costs grow substantially (**Figure 13**). Needing a new wastewater treatment plant, which applies to 34 of 103 project areas, adds a major cost penalty on top of already expensive pipe costs. The most expensive projects combine high pipe density, long conveyance distances, and new treatment plant requirements. Each factor compounds the others.

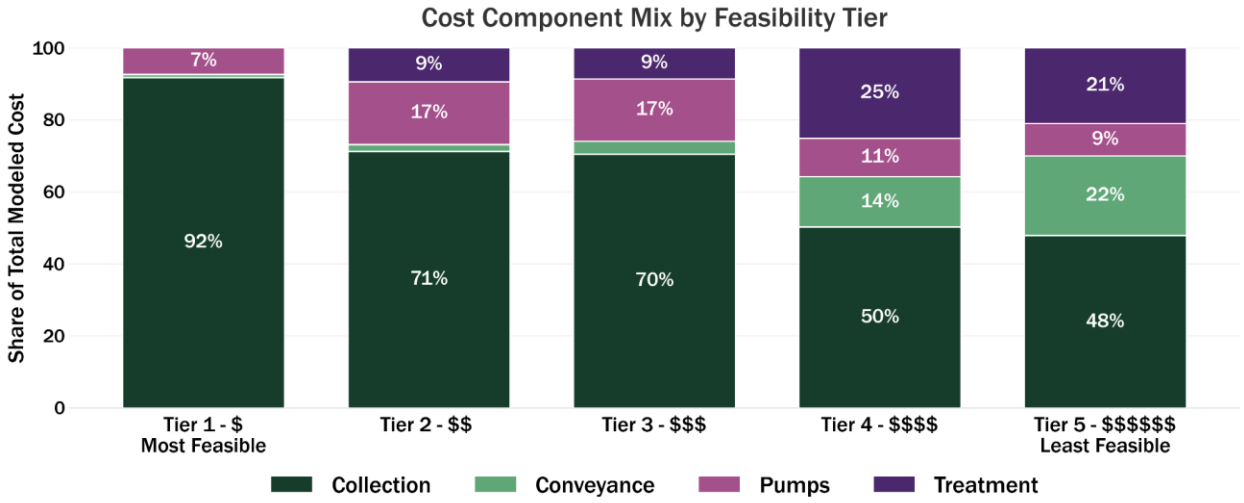


Figure 13. *Share of total planning-level cost by infrastructure component across the five cost feasibility tiers. Collection pipe dominates in the most feasible tiers; treatment plant and conveyance costs grow substantially in Tiers 4 and 5. Areas in the least feasible tiers face cost pressures across multiple components simultaneously.*

These patterns show that the feasibility of municipal-sewer based cesspool replacement is primarily a function of how communities were originally built. Dense, grid-pattern neighborhoods near existing infrastructure will always be less expensive to sewer than dispersed rural subdivisions far from treatment capacity. This means the window for cost-effective sewer expansion is limited to the communities that happen to have been developed at sufficient density, in the right locations, before the 2050 deadline arrives. For the majority of project areas, those conditions were never met.

Lessons Learned From Practitioners and Partners

The key findings above describe what the data collected from planning documents and GIS analysis show, but these data cannot fully capture what experienced professionals know from building, operating, and planning wastewater systems in Hawai‘i. During the study, additional lessons emerged from the stakeholder conversations, public comments, engineering input, review of county and utility planning documents, and the research team’s experience working across agencies and wastewater planning contexts. Together, these sources highlighted practical conditions that shape how wastewater projects are planned, financed, permitted, constructed, operated, and maintained. The observations are presented here as lessons learned rather than results or findings because they are not derived from the quantitative analysis but from the collective experience and expertise of the practitioners, planners, and engineers consulted throughout this study. These ideas are documented below to provide context that data alone cannot capture

Cesspool Conversion Requires Coordinated Action Across State, County, and Utility Partners

Through this study it became clear that technical feasibility is only one part of effective cesspool conversion. Without coordination among the institutions responsible for regulation, funding, infrastructure planning, and implementation, even the most feasible projects will not advance. State law establishes the mandate for conversion, but counties, private wastewater companies, or individual communities carry the primary responsibility for planning, financing, constructing, operating, and maintaining centralized wastewater systems. There is a very real practical challenge of implementing a statewide mandate through infrastructure systems that are locally planned, locally operated, and highly dependent on site-specific conditions. It is clear that counties and private utilities hold essential information that statewide analysis cannot replace, including existing facility constraints, treatment and disposal capacity, planned capital projects, staffing limitations, and community acceptance. Where statewide analysis identifies priority areas, it should be paired with resources, technical assistance, and genuine partnership with local agencies to translate findings into action. A dedicated cesspool conversion program within DOH could provide the institutional continuity needed to coordinate across agencies, track conversion progress, and keep statewide data current as the 2050 deadline approaches. An annual Cesspool Summit could provide a valuable forum for coordination among agencies, lawmakers, counties, private utilities, and other implementation partners. Such a forum could help align policy, funding, infrastructure planning, regulatory implementation, and data-sharing efforts across the statewide cesspool conversion program.

Centralized Sewer Expansion Appears Limited to Infill Areas and Legally Mandated Projects in the Near Term

Through this study it became clear that the most viable near-term candidates for centralized sewer expansion are infill areas adjacent to existing infrastructure, where connection costs are lowest and planning momentum already exists. Legally mandated projects, such as Pāhala and Nāʻālehu on Hawaiʻi Island, represent a separate category where conversion is proceeding regardless of cost-effectiveness. Beyond these two categories, the bar for new centralized sewer expansion is high. County wastewater agencies are largely consumed by the challenge of maintaining and upgrading their existing systems. Aging infrastructure, inflow and infiltration, effluent quality requirements, and deferred capital needs currently absorb the bulk of available wastewater funding. Where cesspool areas are currently being connected to centralized sewer, it is almost exclusively into existing systems with available capacity. The practical implication is that sewer expansion into genuinely new areas competes directly with the urgent and underfunded need to keep existing systems operational, and the data suggest that centralized sewer is not a realistic statewide solution for the majority of cesspools.

Wastewater Disposal Is an Unresolved Constraint on Sewer Expansion

This study did not assess wastewater disposal, what happens to effluent after leaving the WWTP, and that is a significant gap. New ocean outfalls are theoretically possible, but expensive and increasingly difficult to permit. Injection wells, which many Hawaiʻi treatment plants currently rely on, are losing permitting viability and physical capacity over time. Water reuse is likely the only viable long-term pathway, but it is far from straightforward. A treatment plant generating R1 water must include a nitrogen reduction step in its treatment train; without it, spreading treated effluent over non-point areas simply relocates rather than eliminates the nitrogen pollution problem. A sewer project that cannot resolve its disposal pathway cannot move forward regardless of how favorable its collection and conveyance costs appear.

Different Centralized Sewer Technologies Can Obscure the Full Costs of System Ownership

Apparent capital cost savings do not always reflect the full cost of a wastewater system over its lifetime. The choice between gravity and low-pressure sewer illustrates this directly. Low-pressure systems can reduce upfront excavation, pipe, and regional pump station costs, making a project appear more feasible at the screening stage, but they may also distribute pumps, electrical controls, and mechanical equipment across individual properties, shifting long-term maintenance responsibility to homeowners. Counties in Hawaiʻi have consistently expressed preference for gravity sewer because it consolidates maintenance responsibility with trained county staff and avoids placing long-term mechanical obligations on property owners. Decisions made on construction cost alone without accounting for operations, maintenance, and long-term ownership may produce systems that look affordable at ribbon-cutting but become burdensome within a decade. These tradeoffs will certainly arise across technology choices, funding structures, and system configurations, and it is important to weigh lifetime costs and responsibilities, not just construction costs, when assessing feasibility.

Community-Driven Action Has Potential to Advance Cesspool Conversion Where County Planning Has Not

Communities with strong environmental motivation and organizational capacity may be an under-appreciated factor in driving cesspool conversion forward. Example communities in Hawai‘i have secured meaningful progress toward sewer solutions by accessing federal grants, conducting their own technical studies, and building public pressure independent of county planning pipelines. The state could accelerate progress by actively connecting high-priority communities to federal funding pathways and technical assistance rather than relying solely on county capital improvement programs. Community-led efforts in Hanalei, Kaua‘i (Hanalei Initiative, 2025), Mā‘alaea, Maui (Ma‘alaea Village Association, n.d), and Puakō, Hawai‘i Island (Olsen, 2026) demonstrate that this pathway can produce real progress where county planning has not yet reached.

Private Wastewater Providers Could Play a Role but Face Real Economic Barriers

Private wastewater companies already serve several communities in Hawai‘i and could expand their role in cesspool conversion, but the economics are difficult. A private provider's return on investment depends on its rate base, and at costs ranging from three to fourteen times more expensive per connection than a typical onsite conversion, centralized sewer expansion is hard to justify without subsidy or incentive. Subsidized investments may also not count toward the rate base in ways that allow cost recovery through rates, creating a regulatory barrier on top of the financial one. The much higher costs of sewer retrofits as compared to new builds further erodes any return on investment, making these projects unattractive to private capital. Regulatory reform, public-private partnerships, or alternative funding mechanisms may be needed before private providers can practicably take on significant numbers of new cesspool customers.

Climate Change Adds Cost Uncertainty This Study Could Not Quantify

This study did not assess the potential cost impacts of climate change on wastewater infrastructure, though those impacts are real and likely significant. Sea level rise, more intense storms, and changing groundwater conditions could require relocation of existing facilities, adjustment of coastal infrastructure boundaries, and changes to disposal pathways. The HCPT incorporates some climate-related hazard factors, but the cost model does not carry them through. Future analyses should treat climate change as a material cost risk, not a secondary consideration.

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Appendix A: Detailed Cost Analysis Methodology

Cost analysis was performed for each Potential Project Area to estimate cost per cesspool replaced, the primary feasibility metric used in this study.

Disclaimer: The cost model described in this section produces planning-level capital cost approximations that exclude cost of maintenance and operations and are not intended for use in engineering design or for formal project cost estimation.

The aim was to produce comparable, area-specific cost projections that could be evaluated by comparing costs between Potential Project Areas in a relative manner. These projections are then converted into simplified relative cost feasibility tiers for public-facing display on the web tool. The model is designed to be transparent, reproducible, and updateable: as additional project cost data becomes available from county capital improvement programs, engineering studies, and completed construction, the underlying unit costs can be refined and model outputs recalculated without altering the analytical framework.

Cost Model Structure

The cost model estimates total capital expenditure for each Potential Project Area as the sum of four components:

1) Collection Mains
2) Conveyance Mains
3) Pump stations, both neighborhood and regional scale
4) Wastewater treatment capacity needs

Each component is calculated by applying a unit-cost rate derived from Hawaiʻi-based sources such as approved contracts, preliminary engineering reports, and detailed applications for the Clean Water State Revolving Fund. The unit costs are then applied to the measured or estimated physical characteristics relevant to each component type derived from GIS analysis of each Potential Project Area and its relationship to existing infrastructure, topography, and geology.

Collection Cost represents the fully installed cost of gravity sewer mains required to serve properties within the Potential Project Area neighborhood, including pipe, trenching, bedding, and appurtenances. The physical quantity used is the total length of road network within the Potential Project Area boundary, measured in linear feet from GIS analysis. Road length serves as a proxy for the length of collection main needed because sewer mains are typically routed along road alignments to facilitate construction access and minimize disruption to private property. Collection cost is calculated as the product of collection main length, the unit cost per linear foot, and a complicating factor multiplier based on rock type or terrain, as described below.

Conveyance Cost represents the trunk lines, force mains, or interceptors required to route collected wastewater from the Potential Project Area to the nearest existing treatment facility. The physical quantity used is the distance along the road network from the exit edge of the Potential Project Area to the edge of the nearest existing wastewater treatment plant service area, measured in linear feet derived from a GIS-based network analysis as described below.

Conveyance cost is modeled separately from collection cost because conveyance infrastructure typically requires larger pipe diameters, force main configurations, and associated appurtenances to handle the aggregated flow from the entire Potential Project Area, resulting in a higher unit cost per linear foot.

Pumping Costs account for lift stations or pump stations that would be required to move wastewater through the collection and conveyance system where terrain prevents flow by gravity. Pump costs are categorized at two scales. Neighborhood-scale pump stations serve the collection system within the Potential Project Area. Regional-scale pump stations serve the conveyance system connecting the Potential Project Area to the treatment facility. The number and scale of pump stations required for each Potential Project Area is derived from the GIS-based network analysis described below, which identifies locations where terrain causes accumulation points likely requiring pumping. Pump station estimation is among the most uncertain components of the cost model; GIS-based network analysis tends to overestimate the number of required stations relative to what an engineer would determine through field investigation, and the results were calibrated accordingly. This uncertainty is one reason cost results are presented as relative tiers rather than absolute figures.

Wastewater Treatment Capacity Cost accounts for the capital expenditure associated with providing treatment capacity for the additional flow from the Potential Project Area. Where additional treatment capacity is required, costs are estimated the same way whether that capacity comes from a new facility or an upgrade to an existing one. Unit costs derived from Hawai‘i-based sources for both scenarios were similar enough that distinguishing between them was not warranted given the planning-level uncertainty in the nature and scope of either a new plant or an upgrade. When treatment is required, the additional cost is estimated based on the number of cesspools in the project area and three size tiers reflecting documented economies of scale in wastewater treatment plant construction: small facilities serving up to 1,000 connections, medium facilities serving 1,001 to 10,000 connections, and large facilities serving more than 10,000 connections. Treatment cost is calculated as the product of the number of connections in the Potential Project Area and the per-connection treatment rate for the applicable size tier. Where the nearest existing treatment facility has sufficient available capacity to accommodate the projected additional flow, treatment cost is zero. Whether existing capacity is available is based on the best data available at the time of analysis and has not been independently verified for all facilities; this is among the most consequential assumptions in the model, as a change in capacity status directly affects whether treatment cost is zero or substantial.

The total planning cost for each Potential Project Area is the sum of collection, conveyance, pump, and treatment costs. Cost per cesspool replaced, equivalently cost per connection, is calculated by dividing total cost by the number of cesspool connections in the Potential Project Area. The cost model is implemented as a three-tab spreadsheet available at: (https://github.com/cshuler/WW_Overlay_2024/tree/main/Data). Source cost data from all reference projects is compiled and summarized in the ***CostModel_Data tab.*** Unit cost parameters derived from the source data are maintained in the ***CostModel_Parameters tab***. Per-study-area calculations are performed in the ***Costs tab,*** where unit cost parameters from the ***CostModel_Parameters tab*** are applied to GIS-derived quantities for each Potential Project Area. The four component costs are summed to produce total planning cost, which is then divided by cesspool count to yield cost per connection. Cost per connection is then translated

into the cost feasibility tier assignments and the Median Relative Cost (MRC) as described below in order to avoid over-interpretation of the planning-level cost estimates produced through this method.

Unit Cost Derivation

Unit cost rates for each component were derived from documented wastewater project cost data in Hawai‘i. Data sources included Carollo Engineers draft cesspool conversion evaluation report data (Carollo Engineers, Inc., 2026); the AECOM Puna Wastewater Facilities Plan (AECOM Technical Services, Inc., 2023); Maui County FY2026 Capital Improvement Program materials; public bid and contract data for Hawai‘i wastewater infrastructure projects; City and County of Honolulu secondary treatment upgrade documentation for Sand Island and Honouliuli; and State of Hawai‘i Clean Water State Revolving Fund materials. Note that Carollo Engineers draft cesspool conversion evaluation report data is at the time of this report, unpublished data being prepared under EPA’s Closing America’s Wastewater Access Gap Initiative, which provided free technical assistance to Maui, Kaua‘i, and City and County of Honolulu to better understand where cesspool conversion projects could be undertaken and what federal funding sources were available (Carollo Engineers, Inc., 2026). Source costs predating 2022 were escalated to a common dollar year using the Engineering News-Record Construction Cost Index prior to averaging. Carollo's draft studies provided the largest single source of project-level cost data. Because this dataset covers multiple projects across all islands, individual Carollo observations were averaged into a single benchmark before being combined with other sources to prevent the Carollo dataset from having disproportionate influence on the derived unit costs. All source data, dollar years, escalation factors, and derived unit costs are recorded in the CostModel_Data tab of the cost model workbook. Active unit cost parameters were established by reviewing all source observations in the CostModel_Data tab, screening out non-representative projects, and averaging the remaining observations by cost component; current values are shown in **Table A1.**

Collection cost per linear foot is based on a weighted average of the Carollo benchmark, AECOM's published Hawai‘i Island gravity main unit cost, and Maui County sewer extension data. Conveyance cost per linear foot is derived from force main, trunk main, and relief sewer observations drawn from Carollo, AECOM, public bid records, Maui County CIP materials, and Honolulu contract data. Pump station costs are separated into neighborhood and regional scale parameters. The regional pump station value is derived from documented station-specific project costs. Treatment plant unit cost assumptions were developed from Clean Watersheds Needs Survey planning-level cost data (U.S. Environmental Protection Agency, 2012) and adjusted for Hawai‘i construction conditions where appropriate as local pump station data was limited to modification projects not considered representative of a typical new street-scale installation. Treatment costs are modeled in three size tiers based on documented economies of scale in wastewater treatment plant construction. Capacity-based project costs are converted to a per-connection basis using a convention of 725 gallons per day per connection. Small-tier treatment costs are based on decentralized and cluster-scale treatment observations. Medium-tier treatment costs include Central Maui, adjusted AECOM Puna subregional treatment, and Hilo treatment upgrade benchmarks. Large-tier treatment costs are based on large secondary-

treatment upgrade benchmarks from Sand Island and Honouliuli. All data and calculations are publicly available inside of the project data spreadsheet available at (https://github.com/cshuler/WW_Overlay_2024/tree/main/Data).

Table A1. *Active unit costs used in the cost model as derived from multiple data sources*

Parameter	Value	Units	Basis
Collection cost per LF	$1,902	$/LF	Derived Value
Conveyance cost per LF	$3,302	$/LF	Derived Value
Treatment, small (<=1,000 connections)	$146,913	$/conn.	Derived Value
Treatment, medium (1,001 to 10,000 connections)	$72,751	$/conn.	Derived Value
Treatment, large (>10,000 connections)	$18,668	$/conn.	Derived Value
Pump, neighborhood	$750,000	$/each	Planning Value
Pump, regional	$6,533,333	$/each	Derived Value

Construction Cost Adjustments for Difficult Slopes and Basalt Rock Terrain

Excavating through hard volcanic rock and working on steep slopes both increase sewer construction costs substantially above typical conditions. Basalt and other hard volcanic rock require specialized equipment and significantly more time to excavate than softer soils, driving up the cost of trenching for pipe installation. Steep terrain adds complexity to construction access, pipe alignment, and spoil management. Because Hawai‘i’s geology and topography vary significantly across the state, applying a flat per-linear-foot unit cost would systematically underestimate costs in areas with these challenging conditions. Two site-specific adjustment factors were developed in consultation with private and county engineers to account for these realities.

Mean terrain slope for each Potential Project Area was derived from the USGS 3D Elevation Program 30-meter digital elevation model (U.S. Geological Survey, 2024) by calculating slope across the elevation surface and averaging the resulting values within each Potential Project Area boundary. A slope cost factor of 1.10, representing a 10% cost premium, was applied where mean terrain grade is 10 percent or greater. The fraction of each Potential Project Area underlain by basalt and hard volcanic rock was estimated by intersecting statewide geologic map polygons from the geologic map database for the State of Hawai‘i (Sherrod et al., 2021) with each Potential Project Area boundary. The default basalt rock premium is 1.20 where the entire area is underlain by hard volcanic rock, scaled proportionally where only part of the area

is affected; for example, an area with 50% mapped basalt and lava rock receives a 1.10 multiplier. Where both conditions are present, the factors are applied multiplicatively, with a maximum combined factor of 1.32. These adjustments apply to collection and conveyance costs only and by their nature are approximate; they can be reviewed and updated as site-specific information becomes available.

GIS-Based Derivation of Project Area Physical Characteristics

The cost model requires area-specific physical inputs for each Potential Project Area. These are quantities that must be measured or estimated individually, primarily through geospatial analysis of each area. Four key inputs were derived through GIS analysis for each of the 103 Potential Project Areas: collection pipe length, conveyance distance to the nearest treatment facility, estimated number of pump stations, and treatment plant capacity status. The first three were derived through a GIS-based road network analysis described below. Treatment capacity status was determined through the data collection and stakeholder engagement process described in the Results and Key Findings.

Collection pipe length for each Potential Project Area was estimated as the total length of the road network within the area boundary, measured in linear feet using the HI_Z_Simple layer from the Hawaii_Roads_with_Elevation road-centerline dataset, a derived statewide road layer created from Hawai‘i Statewide GIS Program geoportal data (Hawai‘i Statewide GIS Program, 2026). Road length serves as a proxy for collection sewer length because gravity sewer mains are typically installed within public rights-of-way to facilitate construction access and minimize disruption to private property.

Conveyance distance was estimated as the shortest path along the road network from a manually placed neighborhood collection point to the nearest existing wastewater treatment facility or active sewer main endpoint, including both public and private facilities, using a closest facility network analysis in ArcGIS Pro with no slope restriction. The conveyance routing workflow is illustrated in **Figure A1.**

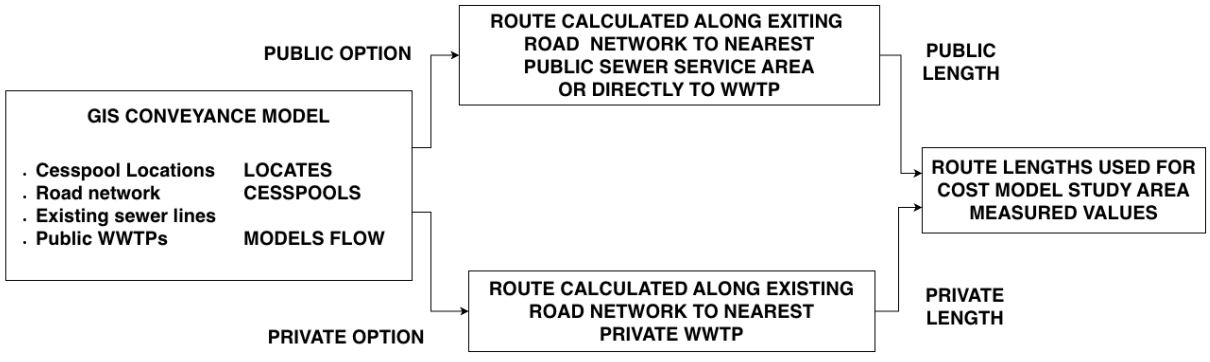


Figure A1*. Simplified workflow for the GIS-based conveyance routing using GIS network analysis.*

Neighborhood pump station counts were estimated using a slope-based directional network analysis in ArcGIS Pro by combining the road network referenced above with statewide Digital Elevation Model (DEM) data. Elevation values from USGS 10-meter Digital Elevation Model datasets for all islands, distributed by PacIOOS (U.S. Geological Survey, 2015), were assigned to road segment endpoints extracted from the statewide road network. Slope was calculated for each segment from start and end elevations. To enable directional flow restrictions, each road segment was duplicated and flipped, with slope values adjusted accordingly, producing a network where each direction of travel carries an independent slope attribute.

Gravity-served parcels were identified by routing parcel centroids to the nearest major road endpoint using a closest facility analysis restricted to segments with negative slope or up to 1% positive slope. This 1% threshold reflects the planning-level assumption that a gravity sewer pipe can be installed at adequate grade beneath a road surface with up to 1% rise; it is not an engineering-level determination and represents one of the more consequential sources of uncertainty in the pump station estimates. Major road endpoints were identified using the Federal HPMS road dataset (Federal Highway Administration, 2025) in ArcGIS Pro as a selector, retaining road network endpoints falling within 10 meters of an HPMS feature. Parcels that could not be routed to a major road under this restriction were classified as non-gravity-served. To identify potential pump station locations, network nodes with no strictly downhill outlet were identified as sink points. Non-gravity parcels were then routed to the nearest sink, and sinks receiving flow were flagged as potential neighborhood pump station locations. The neighborhood collection and pump station estimation workflow is illustrated in **Figure A2.**

As noted above, neighborhood pump station counts derived from this method likely overestimate what an engineer would determine through field investigation. To calibrate the modeled counts, results were compared against the AECOM Puna Wastewater Master Plan (AECOM Technical Services, Inc., 2023), which included approximately one neighborhood pump station for every 1.6 pump stations identified by the network analysis model. Modeled neighborhood pump counts were therefore multiplied by 0.625, reducing counts by 37.5 percent statewide.

Regional pump station counts were estimated by applying the ratio of regional to neighborhood pump stations from the AECOM Puna Wastewater Facilities Plan (AECOM Technical Services, Inc., 2023) to the calibrated neighborhood pump count for each Potential Project Area, yielding approximately one regional pump station for every 2.7 neighborhood pump stations. This ratio was applied to the calibrated neighborhood pump count for each Potential Project Area to estimate regional pump station requirements. It should be noted that no locations for regional pump stations were determined, only estimated numbers were calculated. These estimates are not intended to represent final hydraulic design but provide a consistent planning-level method for comparing relative pumping requirements and associated costs across Potential Project Areas.

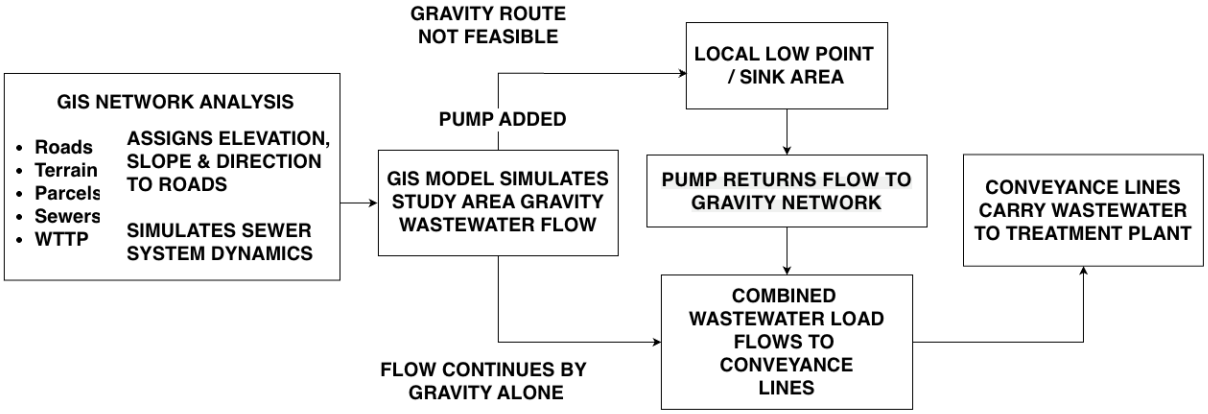


Figure A2. *Simplified workflow for the GIS-based slope-directed network analysis used to estimate neighborhood pump station locations and counts.*

Treatment plant capacity status for each Potential Project Area was determined through direct consultation with county wastewater departments and supplemented by review of published planning documents, facility master plans, and PUC filings for private utilities. For each area, capacity status was assigned as available, needed, or unknown based on the best information obtainable at the time of analysis. County-operated systems were generally better documented; private utility capacity data was less consistently available and more frequently assigned an unknown status. As noted in the cost model structure section, capacity status is among the most consequential assumptions in the model. Areas with WWTPs with enough available capacity to serve the number of cesspools in the Potential Project Area were assigned zero treatment cost. Again it should be recognized that any corrections to a WWTPs capacity status could add or subtract substantial cost and may shift the cost feasibility tier assignment for the areas served.

Cost Feasibility Tiers

The web tool displays cost information as relative cost feasibility tiers rather than specific dollar estimates. This approach was adopted deliberately to reflect the planning-level accuracy of the underlying cost model and to prevent users from interpreting modeled estimates as engineering-grade project costs. Planning-level cost estimates of this type are generally characterized by the Association for the Advancement of Cost Engineering as Class 5, carrying an expected accuracy range of -50% to +100% of actual project costs. The estimates produced by this model should be interpreted with uncertainty equal to or greater than this range. Key inputs including collection pipe length, conveyance distance, and pump station counts are derived from GIS proxies rather than field investigation, and treatment plant capacity status remains unverified for several areas. For these reasons, cost results are presented as relative tiers rather than absolute figures, as the comparative rankings between areas are more meaningful than the underlying dollar values. With this in mind cost per connection values were translated into five relative cost feasibility tiers, as shown in **Table A2**. The ratings are public-facing planning categories, not engineering estimates or funding commitments.

For analyses requiring more granular cost comparison than the five-tier system provides, cost is also expressed as the Median Relative Cost (MRC), a dimensionless value calculated as each project area's cost per connection divided by the statewide median cost per connection. An MRC of 1.0 represents the median project; an area at 0.5 costs half as much per connection as the typical project, and an area at 2.0 costs twice as much.

Table A2. *Relative cost feasibility tiers and MRC range applied to all 103 Potential Project Areas.*

Rating	Cost per Connection	Median Relative Cost (MRC) Range	Approx. Multiple of Onsite Alternative	Areas	Share
(1) - $	Under $120000	Under 0.3	Under 3x	4	4%
(2) - $$	$120,000-$200,000	0.3–0.5	3x-5x	7	7%
(3) - $$$	$200,000-$320,000	0.5–0.8	5x-8x	26	25%
(4) - $$$$	$320,000-$560,000	0.8–1.4	8x-14x	44	43%
(5) - $$$$$	Over $560000	Over 1.4	Over 14x	22	21%

[1] Onsite system cost assumed at $40,000 per connection as a planning-level baseline as reported in Hawai'i (Ishida, 2023; Azambuja, 2024).

Cost Model Consistency Check Against Carollo Project Data Costs

To evaluate the internal consistency of the cost model, modeled cost per connection estimates were compared against bundled project-total cost data from the Carollo Engineers draft integrated wastewater management plans (Carollo Engineers, Inc., 2026). These bundled project totals represent the same projects whose component-level costs were used to derive unit cost parameters, so this comparison is best understood as a consistency check rather than an independent validation. It checks whether applying this study's blended statewide unit costs back to the source projects after integration of other cost model input datasets yields Potential Project Area costs within a reasonable range consistent with the counties' own management plans.

Across the three infrastructure configurations where comparison data were available, modeled estimates ranged from 7 to 28 percent above Carollo project averages (**Table A3**). A significant portion of this difference is attributable to a deliberate scope difference rather than model error: Carollo's designs incorporated a mix of gravity and low-pressure sewer collection, with approximately 41% of connections in collection-only projects. The cost model, by contrast, assumes 100% gravity sewer throughout, a methodological choice made for consistency across all project areas and reflecting county preferences for gravity systems as described in the methods section. When modeled costs are scaled to account for this offset, the adjusted estimates track Carollo's bundled project costs more closely, although substantial project-level

variability remains (**Figure A3**). This variability reflects genuine uncertainty at the project level rather than systematic bias, and reinforces the importance of treating these as planning-level estimates rather than engineering-grade projections. No Carollo bundled project data were available for the most complex configuration with all cost model components, which carries the highest modeled costs and likely the greatest uncertainty. These results are broadly consistent with a planning-level tool and suggest the model performs well and tracks reasonably with the Carollo project costs. Importantly, the cost model is conservative in that it does not systematically underestimate costs, though the degree of conservatism varies by project type.

Table A3. *Cost model consistency check comparing average modeled cost per connection to average observed project cost per connection from Carollo Engineers, Inc. (2026), by infrastructure configuration.*

Infrastructure Configuration	Carollo Bundled Project Cost/Connection	Cost Modeled Total Cost/Connection
Collection only	$138,733	$177,632
Collection + pump stations + force main	$306,474	$327,215
Collection + new WWTP	$344,367	$383,950
Collection, conveyance + WWTPs + pumps	NA	$464,857

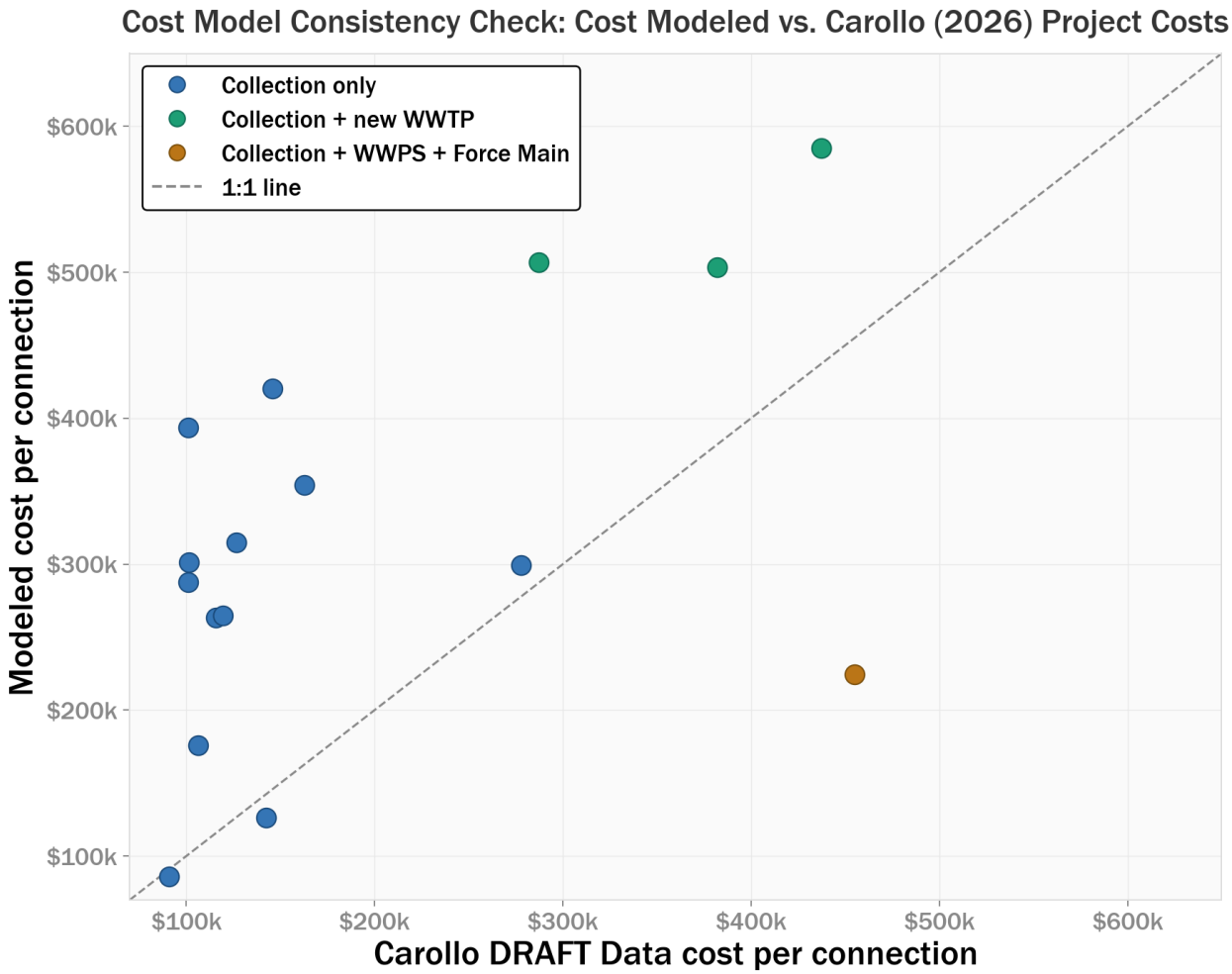


Figure A3. *Project-level comparison of Carollo Engineers estimated cost per connection versus modeled cost per connection for the 15 Potential Project Areas present in both datasets. Points above the 1:1 line indicate this study's cost model estimates higher costs than Carollo Engineers, Inc., (2026). The systematic offset reflects in part that Carollo's designs incorporated an average of 41% low-pressure sewer connections, while the cost model assumes 100% gravity sewer throughout. Color indicates infrastructure configuration as classified by the validation category framework.*

Additional Validation: Last-Minute Cost Comparison with Concurrent Carollo Engineers Study

This report was prepared in response to Act 217 and accordingly is due to the Hawai‘i State Legislature on June 30, 2026. Throughout the project period, the study team interfaced regularly with Carollo Engineers, Inc., who through a notable coincidence received federal technical assistance funding from the U.S. EPA under the Closing America's Wastewater Access Gap Initiative to conduct county-level cesspool conversion evaluations across Hawai‘i at approximately the same time this Legislature-mandated study was underway. While the scope of these efforts was not entirely duplicative, both projects were examining the viability of

centralized sewer as a pathway for cesspool replacement. The Carollo studies were conducted at the county level and were oriented toward planning-level engineering evaluation of specific project areas, whereas this study focused on integrating broader statewide datasets and developing a publicly accessible tool for querying and exploring cesspool conversion feasibility across the state.

The study team and Carollo Engineers agreed early in the project to align methods where possible, and a number of the methodological choices reflected in this report, including the project area delineation approach, were informed directly by Carollo's work. The authors of this report are grateful for that collaboration and for Carollo's willingness to advise on technical matters throughout the project period.

Unfortunately, the timing of the two efforts did not align well. Carollo's county-level reports were not finalized or released prior to the June 30, 2026 deadline for this report. Carollo did share preliminary cost model data with the study team at an earlier stage, which was used to develop and calibrate the cost model structure described in Appendix A and the consistency check presented there. The full Carollo reports remain in draft and have not been publicly released as of this writing. When they are released, it would be worthwhile to revisit the Act 217 Sewer Expansion Overlay Tool dataset in light of the new project area delineations and cost data, though doing so in a meaningful way would likely require additional time and funding beyond the current scope.

Four days prior to the submission deadline for this report, the study team received an advance draft of the Carollo cesspool conversion evaluation for the City and County of Honolulu. While the structure of this study, its cost model, and its project area delineations could not be revised at that stage, Carollo’s Honolulu data provided an opportunity to conduct an independent cost validation for Oʻahu project areas, the results of which are presented below.

Validation Methodology: The Carollo report independently identified 16 potential sewer project areas on Oʻahu and developed planning-level cost estimates for each. Because these areas overlap substantially with 16 of the 22 Potential Project Areas identified by this study on Oʻahu, the two datasets provide an opportunity for an independent cost comparison.

Project area boundaries were matched by map location, name and cesspool count. Where Carollo consolidated multiple density clusters into a single project area, the corresponding Potential Project Areas were merged and a weighted average cost per connection was calculated. Cesspool counts between the two studies are generally within 10 percent, reflecting minor differences in boundary delineation and inventory vintage rather than methodological disagreement.

Figure A4 plots the modeled cost per connection from this study against Carollo's DRAFT planning-level estimates for all 16 matched areas. The two datasets show a positive relationship ($R^2 = 0.54$), indicating the cost model captures the same relative cost patterns that Carollo's independent analysis identified. The regression slope of approximately 0.70 with a positive intercept suggests this study’s model tends to overestimate costs for lower-cost projects and underestimate costs for higher-cost projects relative to Carollo's estimates. This pattern is consistent with the consistency check above and the gravity-sewer-only assumption used throughout this study: Carollo's designs incorporated a mix of gravity and low-pressure sewer

collection, and LPS connections carry lower capital costs per connection than gravity. Projects where Carollo relied heavily on LPS, generally lower-cost infill areas, will appear above the 1:1 line in **Figure A4**, while projects where both studies assumed gravity-dominant collection track more closely.

Overall, this validation opportunity reinforces the conclusions in the Consistency Check section above, the cost model performs reasonably well as a relative ranking tool at the planning level, and the systematic offset reflects a documented scope difference rather than model error. Cost feasibility tier assignments should be interpreted accordingly.

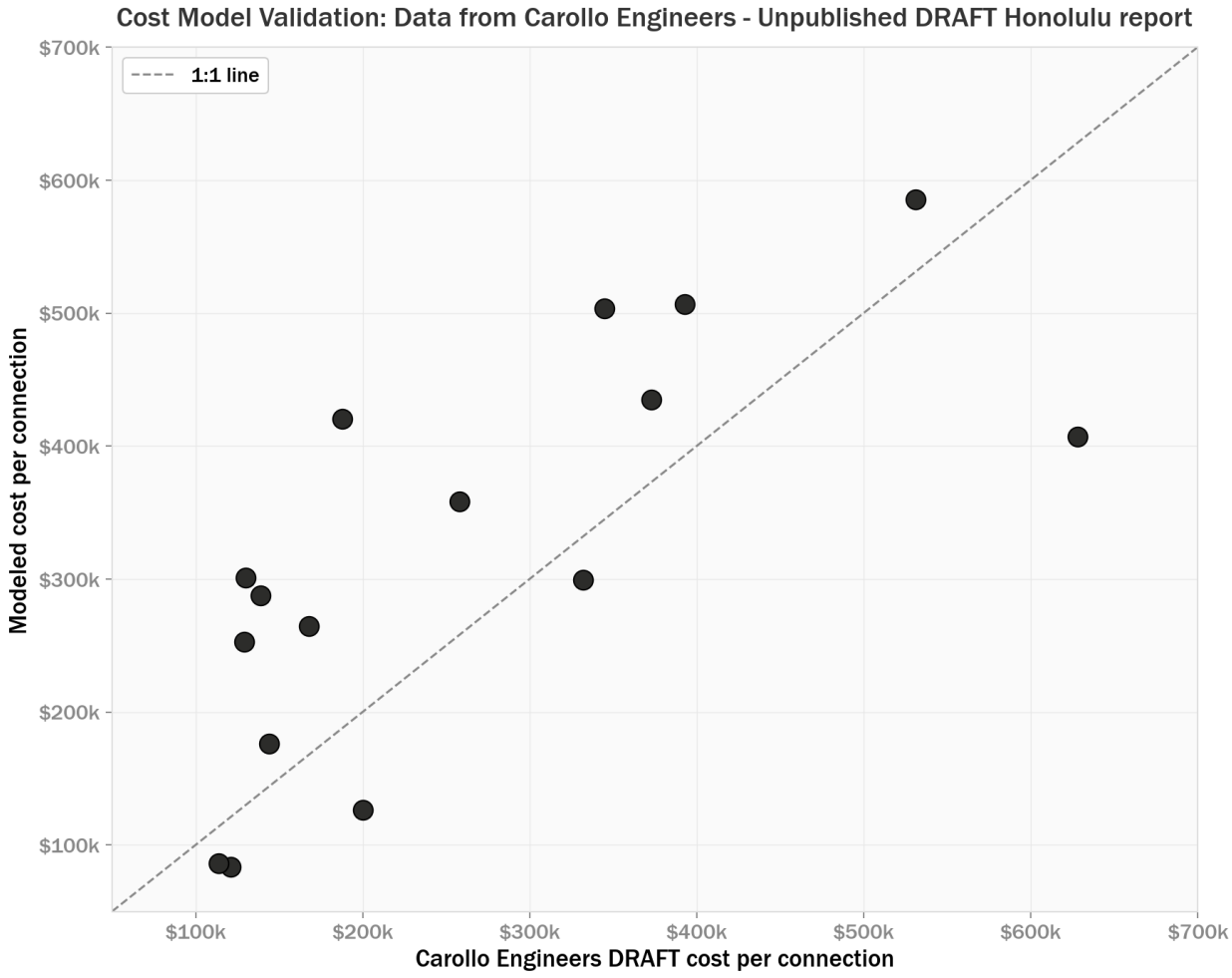


Figure A4. *Project-level comparison of Carollo Engineers estimated cost per connection versus modeled cost per connection for 16 Potential Project Areas on Oʻahu present in both datasets. Points above the 1:1 line indicate this study's cost model estimates higher costs than Carollo Engineers, Inc. (2026). The systematic offset reflects in part that Carollo's designs incorporated a mix of gravity and low-pressure sewer connections, while the cost model assumes 100% gravity sewer throughout.* $R^2 = 0.54$.

Cost Model Limitations and Assumptions

The cost model estimates capital expenditures only. Operating and maintenance costs are intentionally not modeled because operating cost structures differ significantly between system types and jurisdictions and cannot be estimated meaningfully at the planning level without detailed engineering analysis.

The model has not been validated against actual completed project costs. All unit costs are derived from planning documents, draft studies, and contract data rather than final as-built costs. The degree to which modeled estimates track actual construction outcomes is unknown. This is the most fundamental limitation of the model and the primary reason cost results are presented as relative tiers rather than absolute figures.

WWTP capacity status is among the most consequential and least certain inputs in the model. Where existing capacity is assumed available, treatment cost is zero; a county correction to that status would add substantial cost and could shift the cost feasibility tier assignment for that area. Capacity data for county-operated systems was obtained through direct consultation and is reasonably current. Capacity data for privately operated systems was based on PUC filings and direct outreach where possible and is less reliably verified. Several areas retain an unknown capacity status, and their cost feasibility tier assignments should be interpreted accordingly.

Pump station counts are derived from GIS-based network analysis and may overestimate the number of stations an engineer would identify through field investigation, as described in the network analysis section. Results were calibrated to reduce this bias but residual overestimation could remain and contribute to cost uncertainty particularly in areas with complex terrain.

Disposal costs are not captured in the model. This omission is significant for areas proximate to treatment plants operating under injection well permits rather than NPDES ocean outfall permits. Following the functional equivalence doctrine established in County of Maui v. Hawai‘i Wildlife Fund (2020), such plants may face regulatory constraints requiring alternative disposal infrastructure at costs that could substantially exceed the collection and conveyance costs modeled here. This constraint has been identified as a potential barrier to sewer expansion in areas served by the Pa‘alakai, Kahuku, and Waimānalo treatment plants on O‘ahu. Cost feasibility tiers for these areas should be interpreted with this caveat in mind.

Some unit cost parameters, particularly neighborhood pump station costs, are based on limited data points and should be read as planning-level assumptions rather than estimates derived from large samples. The model is structured to incorporate new source data as it becomes available and to recalculate derived parameters and area-specific outputs accordingly.

Appendix B : Definitions and Commonly Used Terms

This glossary provides working definitions for terms used in this report. Where a term has a regulatory basis, wording is aligned with the relevant Acts and administrative rules. Where a term is project-specific or in common use, the definition reflects how it is used in this report and its supporting documents.

Act 125 (2017): The 2017 Hawai‘i law that established the statewide requirement that cesspools be upgraded, converted to an approved onsite system, or connected to a sewerage system before January 1, 2050, unless exempted by the Director of Health. In this report, Act 125 supplies the broad compliance deadline and the statewide mandate that frames the need for replacement pathways.

Act 132 (2018): The 2018 Hawai‘i law that established the Cesspool Conversion Working Group and directed development of a long-range statewide cesspool conversion plan. In this report, Act 132 is part of the planning lineage between Act 125's statewide mandate and Act 217's more specific request for an overlay identifying where sewer or other centralized treatment may be feasible.

Act 217 (2024): The 2024 Hawai‘i law that directs the University of Hawai‘i Water Resources Research Center and the University of Hawai‘i Sea Grant College Program to develop an overlay for the Hawai‘i Cesspool Prioritization Tool. The overlay is intended to identify areas where a county sewer system or other centralized treatment system may be feasibly expanded or constructed to reduce or eliminate cesspools before January 1, 2050. Act 217 identifies relevant feasibility factors, including development density, proximity to existing wastewater infrastructure, county plans, timing, cost, and other considerations.

Aerobic Treatment Unit (ATU): Onsite wastewater treatment technologies that, unlike standard septic systems, use aeration to support biological treatment before discharge. In this report, ATU refers to both NSF/ANSI Standard 40 systems, which provide conventional secondary treatment, and NSF/ANSI Standard 245 systems, which provide enhanced nitrogen removal. Both are commonly referred to as ATUs in practice, in HAR 11-62, and throughout this report. Under Act 125, ATU conversion is one approved pathway for cesspool replacement where sewer connection is not feasible, where a higher level of treatment is required by regulation or site conditions, or where it is selected at the property owner's discretion.

Capital Cost (CapEx): The one-time costs of designing, permitting, and constructing wastewater infrastructure, borne by counties, private utilities, or property owners depending on the system. The cost model used in this report quantifies planning-level capital costs directly, making those the primary component of the analysis; OPEX (defined below) is addressed mainly as a limitation.

Cost Model: The planning-level tool used in this project to estimate relative infrastructure costs and assign cost feasibility tiers that factor into feasibility evaluation across Potential Project Areas. It is based on a composite of cost data from centralized wastewater infrastructure projects in Hawai‘i from 2019 to 2026, drawn from capital improvement programs, engineering reports, and planning studies. These data are used to develop averaged unit costs applied to measured system elements within each Potential Project Area. All outputs are high-level

planning estimates, not actual construction cost projections, and do not include long-term operation and maintenance costs unless explicitly stated.

Centralized Sewer Expansion: The extension of an existing sewer system, or the construction of related centralized wastewater infrastructure, to serve an area currently using cesspools or other onsite systems. In this report, centralized sewer expansion refers primarily to the physical expansion of collection and conveyance infrastructure. Where data are available, the term also considers whether adequate treatment capacity exists to receive and manage additional wastewater flows.

Centralized Treatment System: A wastewater system serving multiple properties through shared infrastructure, generally including collection, conveyance, treatment, and disposal or reuse components. The term includes county sewer systems and may also include private or utility-scale systems where they function as shared treatment infrastructure for a defined service area. Also see Wastewater Treatment Works (WWTW).

Cesspool: An individual wastewater system consisting of an excavation in the ground that receives untreated wastewater and permits liquid to seep through the bottom or sides into the surrounding subsoil with little to no treatment, where it may reach groundwater and contaminate drinking water sources and nearshore ecosystems. Cesspools represent an outdated technology that has been phased out or banned in most U.S. jurisdictions. In this report, cesspools are the primary inventory unit targeted for replacement or connection.

Cesspool Cluster / Potential Project Area: A mapped area containing a concentration of cesspools or other onsite wastewater disposal systems that meets the project's density and proximity screening criteria, making it a candidate for evaluation as a potential centralized service area. In the current methodology, a cluster is generally defined as 50 or more Class IV OSDS (cesspools) located within one-eighth of a mile of one another and within two miles of an existing wastewater treatment facility. Qualifying clusters are refined into mapped polygons and used as the primary unit of analysis throughout the report.

Cesspool Conversion: The process of eliminating a cesspool by upgrading or converting it to an approved onsite wastewater system such as a septic tank or ATU, or connecting it to a sewerage or centralized treatment system. In this report, conversion is used broadly and does not imply that sewer connection is always the preferred or feasible solution.

Collection System and Conveyance: The infrastructure used to gather and move wastewater from individual properties to a treatment facility. The collection system includes the local network of street sewer laterals, gravity mains, pump stations, and force mains that serve a neighborhood or Potential Project Area. Conveyance refers to the larger trunk lines, interceptors, and force mains that carry aggregated flow from the Potential Project Area to an existing or proposed treatment facility. In the cost model, collection and conveyance are estimated separately because conveyance infrastructure is typically larger, longer, and more complex, and may require pump stations or force main configurations that add capital cost.

Cost Feasibility Tier: A classification system consisting of five relative cost categories (ranging from lowest to highest) used to evaluate and compare the financial viability of centralized sewer expansion across Potential Project Areas. Because underlying cost model estimates carry significant planning-level uncertainty, these tiers are used instead of absolute dollar figures and

are expressed as approximate multiples of a typical onsite cesspool-to-septic conversion cost to provide an intuitive sense of scale.

Cost Model: A planning-level analytical framework used to estimate the total capital cost of extending centralized sewer infrastructure to a Potential Project Area. Calculated on a per-cesspool basis to allow for direct comparison, the model sums the estimated costs of four core infrastructure components—collection mains, conveyance mains, pump stations, and wastewater treatment capacity—by applying documented regional unit cost rates to physical quantities measured through GIS analysis. Due to inherent planning-level uncertainties, the model expresses its outputs as five relative cost feasibility tiers rather than absolute dollar figures.

County Sewer System: A centralized wastewater system owned, operated, and maintained by a county wastewater agency. In Hawai‘i, public wastewater systems are generally county-owned and operated, with private systems serving a limited number of smaller communities, large condominiums, and resorts. In this report, county sewer systems are the primary reference point for existing service areas, planned expansion, treatment capacity, and institutional feasibility.

Decentralized Wastewater System: A wastewater treatment and disposal approach typically serving a small number of neighboring properties, independent of county-owned or regional sewer networks. In this report, decentralized systems are most relevant for clusters that are distant from existing infrastructure, outside documented plans, or unlikely to be cost-effective for centralized expansion but that contain sufficient density to warrant consideration.

Density-Based Clustering Analysis: The method used to identify concentrations of cesspools suitable for centralized sewer expansion feasibility screening. Geographic Information Systems (GIS) are used to analyze spatial relationships between parcels containing cesspools to identify those meeting the selected density thresholds. Qualifying parcels are grouped into clusters and refined into candidate Potential Project Area polygons using roads, topography, natural features, existing sewer infrastructure, and planning judgment. Potential Project Area boundaries were further reviewed and refined through consultation with county wastewater agencies to reflect local planning knowledge and documented infrastructure conditions. Also see Proximity to Wastewater Infrastructure.

Disposal System: A facility or infrastructure component used for the final disposal of treated or partially treated wastewater or wastewater sludge. The term encompasses a range of approaches including sewers, seepage pits, injection wells, soil absorption systems, disposal trenches, and associated transmission lines, pumps, and equipment. In this report, disposal may include subsurface disposal, surface discharge, reuse, or other permitted end pathways depending on the system type.

Effluent: Treated or partially treated wastewater leaving a treatment unit, treatment facility, or disposal-related process. In this report, effluent matters because treatment capacity alone does not determine feasibility; the receiving environment, permitted disposal route, reuse options, water quality requirements, and facility condition may all affect whether additional flows can be accepted.

Effluent Reuse or Recycled Water: The beneficial use of treated wastewater effluent for an approved purpose, subject to applicable water quality and monitoring requirements. In the

report, reuse is relevant because a treatment facility may have hydraulic or treatment capacity on paper but still face practical constraints related to effluent quality, reuse demand, disposal location, or permit conditions. The highest grade of recycled wastewater is R1.

Feasibility: The relative practicality of replacing cesspools through centralized sewer or other centralized treatment infrastructure in a given area, defined for the purposes of this report primarily as a function of cost per cesspool replaced. Feasibility is not a binary condition and reflects a combination of factors including density, proximity to existing infrastructure, county or utility plans, timing, engineering constraints, treatment capacity, and institutional readiness. In addition to cost per cesspool replaced, Potential Project Areas are assigned planning status classifications that distinguish between Planned or Underway (P), Under Evaluation (E), and No Active Plans (N). These classifications are planning tools, not permit decisions, funding commitments, or final engineering determinations.

Force Main: A pressurized pipe used to convey wastewater from a pump station or lift station toward a downstream gravity system, interceptor, or treatment facility. Force mains are part of the conveyance system and can increase project cost in proportion to their length, depth, and substrate conditions.

Gravity Sewer: A type of sewer line that is designed to carry wastewater downhill using gravity rather than continuous pumping. The report treats gravity systems as generally preferred by most counties because they align with existing operations and reduce homeowner-side maintenance concerns, although they can be more expensive or difficult to construct where topography is unfavorable compared to smaller low-pressure systems. Also see Low-Pressure Sewer (LPS).

HAR 11-62: Hawai‘i Administrative Rules Chapter 11-62, Wastewater Systems. In this report, HAR 11-62 provides the regulatory vocabulary and technical context for wastewater systems, treatment works, treatment units, collection systems, disposal systems, individual wastewater systems, and related requirements.

Hawai‘i Cesspool Prioritization Tool (HCPT): A statewide GIS-based tool that evaluates the relative hazard posed by cesspools using environmental and human-use factors, identifying where cesspool impacts are most likely and most urgent. The HCPT does not, by itself, determine how cesspools should be replaced or whether centralized sewer expansion is feasible. This report and its accompanying web tool represent an update and expansion of the HCPT, both of which will receive regular updates as new data and planning information become available.

Individual Wastewater System (IWS): An onsite wastewater system serving an individual property or facility, as distinct from a centralized treatment works. Cesspools, septic systems, aerobic treatment unit systems, and other approved onsite technologies fall within this general replacement-pathway discussion, although each has different regulatory and site suitability requirements.

Lift and Pump Stations: Pumping facilities used to move wastewater where gravity flow is insufficient. In the cost model, pump costs may be assigned at neighborhood scale, regional scale, or both, depending on topography and the system configuration required to connect a Potential Project Area to a treatment facility. Pump locations, quantities, capacities, and resulting cost estimates are all planning-level figures.

Low-Pressure Sewer (LPS): A sewer system that uses grinder pumps or similar equipment to move wastewater through small-diameter pressurized lines. LPS may reduce some capital costs compared with gravity sewer, but it can shift costs and responsibilities into operation, maintenance, equipment replacement, and homeowner-side issues. The report generally standardizes the planning analysis around gravity sewer and force mains unless the local context indicates otherwise.

Operation and Maintenance Expense (O&M or OPEX): The recurring costs and operational responsibilities required to keep wastewater infrastructure functioning after construction, carried out by county or private wastewater entities. The cost model used in this report focuses on capital cost only; O&M is therefore acknowledged as a limitation rather than a quantified component.

Overlay: The project layer developed with the HCPT under Act 217. The overlay adds or "overlays" centralized sewer expansion feasibility information to the existing HCPT hazard framework, including cost feasibility tiers and planning status classifications. It is intended to help identify where centralized service may be practical, already planned, under evaluation, or not currently supported by documented planning activity. It does not replace county planning, DOH permitting, engineering design, or property-specific decision making.

Parcel / Tax Map Key (TMK): The parcel-level spatial unit used in this study to organize and analyze property records, cesspool and OSDS locations, sewer billing information, and potential service relationships. TMKs are useful because they provide unique statewide property identifiers that can be aggregated into larger planning areas for analysis.

Planning Status: A classification used in this report to describe the documented planning or implementation status of potential centralized sewer expansion for a Potential Project Area. Planning status is assigned from available documents, county or utility input, CIP information, and other planning evidence. It does not establish regulatory approval, funding eligibility, construction readiness, or final project feasibility.

Planned or Underway (P): A sewer expansion project that has been formally approved, funded, or scheduled for implementation by the relevant county, wastewater authority, or responsible utility provider. This category includes projects currently under construction or actively being implemented, as well as projects confirmed by the responsible wastewater utility provider to be built within the next 10 years.

Under Evaluation (E): A potential sewer expansion project that has been identified in planning documents, studies, or published capital improvement program (CIP) budgets but has not been formally approved, funded, or scheduled for implementation. Projects in this category are being evaluated or considered as possible future infrastructure investments but remain highly uncertain.

No Active Plans (N): No known sewer expansion projects, studies, or planning efforts currently identify the area for sewer infrastructure expansion. At the time of analysis, no documented plans, proposals, or evaluations related to sewer expansion were found. However, the area contains cesspools at sufficient density that a centralized wastewater system could be a potential future solution.

Planning-Level Estimate: A Class 4 or Class 5 feasibility-level cost estimate, as defined by AACE International, suitable for comparing alternatives and identifying priorities across Potential Project Areas. These estimates are based on measured spatial quantities and documented unit costs from recent Hawai‘i wastewater projects but reflect limited project definition and carry broad expected accuracy ranges. They are not suitable as final engineering estimates, bid documents, or funding commitments, and should be refined as additional project-specific engineering data and reports become available.

Potential Project Area: Mapped areas where cesspool density meets a minimum threshold for centralized gravity sewer to be economically viable or have otherwise been identified as a viable area for inclusion by county staff.

Private Utility Service Area: A wastewater service area associated with a private utility or non-county wastewater provider, often documented through Public Utilities Commission filings or other official records. In the report, these service areas may be relevant where centralized service is possible outside a county-operated sewer system.

Proximity to Wastewater Infrastructure: The spatial relationship between a cesspool cluster or Potential Project Area and existing sewer mains, laterals, service areas, pump stations, or treatment facilities. Proximity is a primary feasibility driver because distance strongly affects the length and cost of required collection and conveyance infrastructure.

Onsite Wastewater Disposal System (OSDS): A broad working term for parcel-scale wastewater systems not connected to centralized sewer, used primarily in the inventory and GIS analysis for this report. OSDS records include cesspools and other onsite system types. The statewide OSDS inventory used in this project was screened, cleaned, and updated as part of the project methodology. See IWS definition above.

Sewer Service Area: A mapped area currently served by, planned to be served by, or associated with a centralized wastewater system. In the report, service areas are used to determine proximity, possible expansion relationships, and whether a cesspool cluster may be connected to existing or planned infrastructure.

Treatment Unit / Wastewater Treatment Plant (WWTP): Under HAR 11-62, a treatment unit is a plant, facility, or equipment used in the treatment of wastewater, including pumps, power equipment, blowers, motors, holding tanks, flow splitters, and other process equipment. In report usage, the WWTP refers to the treatment facility or treatment portion of a broader wastewater system or treatment works.

Treatment Works / Wastewater Treatment Works (WWTW): Wastewater Treatment Works or WWTW is used here to refer to the broader treatment system, not just the plant. This includes the treatment unit and associated collection and disposal systems, excluding individual wastewater systems, consistent with HAR 11-62's definition of treatment works.

Trunk Line / Interceptor: A large-diameter sewer line that receives flow from local collection systems and conveys it toward a treatment facility or major downstream connection, sometimes as a force main under pressure. In the cost model, trunk lines and interceptors are treated as conveyance infrastructure, distinct from neighborhood-scale collection.

Wastewater / Wastewater System: Wastewater refers to any liquid waste, whether treated or untreated, including domestic sewage and other liquid wastes. In this report it most often refers to domestic wastewater from homes and other buildings served by cesspools, onsite systems, or sewer systems. A wastewater system is the broader term encompassing all infrastructure and processes involved in the treatment, use, and disposal of wastewater and wastewater sludge, including treatment works, collection systems, recycled water systems, and individual wastewater systems, consistent with HAR 11-62.

Wastewater Treatment Capacity: The ability of a treatment facility to accept and process additional wastewater flow while meeting operational, permit, effluent quality, and disposal or reuse requirements. In this report, treatment capacity is not limited to design capacity on paper. Plant condition, infiltration and inflow, permit constraints, and effluent disposal or reuse pathways may all affect whether a facility can meaningfully accept new connections from a Potential Project Area.

Wastewater Treatment Capacity Cost: The planning-level capital cost associated with providing treatment capacity for additional flow from one or more Potential Project Areas. Where an existing facility has confirmed available capacity, this cost component may be limited or zero in the model. Where new or expanded treatment capacity would be required, the model assigns cost based on estimated additional flow from the Potential Project Area and an assumed cost per gallon per day of new treatment capacity.